LLMpedia: A Transparent Framework to Materialize an LLM's Encyclopedic Knowledge at Scale

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Abstract

Benchmarks like MMLU suggest flagship language models approach factuality scores above 90%. *LLMpedia* shows this picture is incomplete. We materialize 1.3M encyclopedia articles entirely from parametric memory across three model families, and verify claims against Wikipedia and curated web evidence. For `gpt-5-mini`, true rate is 68.4% on Wikipedia-covered subjects - more than 21 pp below MMLU. The gap is mostly due to *unverified* claims (30.5%), not refutation (1.2%); an audit attributes 90% of them to missing evidence. Wikipedia covers only 56.7% of model-surfaced subjects, and beyond these, true rate drops to 57.6%. In a retrieval-trap benchmark against Grokipedia, LLMpedia is more factual at roughly half the textual similarity to Wikipedia. We measure verifiability, not editorial quality. We release articles, code and data at https://llmpedia.net.

1 Introduction

Hundreds of millions of users now read LLM output as fluent multi-paragraph prose rather than as multiple-choice answers (OpenAI, 2025b; Anthropic, 2026; Chatterji et al., 2025; Handa et al., 2025), yet factuality is still measured in artifical suites like MMLU (Hendrycks et al., 2021), TruthfulQA (Lin et al., 2022), HLE (Phan et al., 2025). These report how often a model answers the questions an experimenter chose to ask, and adding more questions does not close the gap, because question sets are drawn from subjects that reference works already document: the regions where a model's knowledge is thinnest are the regions least likely to be tested (Hu et al., 2025). This is the *availability bias* of Tversky and Kahneman (1973), and its implications are significant. Benchmarks systematically overestimate the factual reliability of parametric knowledge in the long-form modality users read (Saeed et al., 2025a), because they evaluate on what benchmark designers ask.

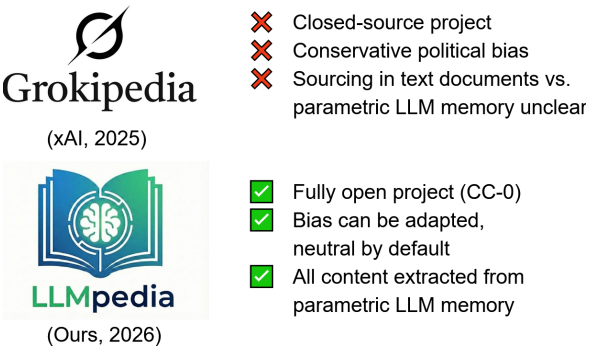


Figure 1: LLMpedia generates entirely from parametric memory with full auditability. Grokipedia's opaque pipeline shows evidence of retrieval-shaped generation (Yasseri and Mohammadi, 2025).

Understanding that gap requires a different probe rather than a better benchmark, one in which the model and not the experimenter chooses the subject. *Recursive materialization* works this way: we ask a model to write about a subject, take the entities it mentions, and write about those in turn, so the resulting corpus reflects what the model is willing to assert. Cohen et al. (2023) introduced recursive elicitation and Hu et al. (2025) scaled it into GPTKB (100M+ triples) with availability bias as the explicit target, but triples are not the form most users consume (Saeed et al., 2025a), so we materialize into encyclopedia articles instead. The article is a medium here and not a product: LLMpedia does not aim to replace or compete with Wikipedia. We choose it because Wikitext is close to the prose readers see, and because Wikipedia then gives a reference against which both those claims and the subjects themselves can be compared.

The last of these is what makes the evaluation work. We decompose generated articles into

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atomic claims and label samples against retrieved evidence as *supported*, *refuted*, or *insufficient*, taking that evidence from the Wikipedia article on the subject and, where none exists, from sources meeting a quality rule under which government and academic domains qualify and the ∼133 domains we score by hand are examples rather than the limit (§4.2). The second tier matters because Wikipedia itself skews Western (Wikipedia, 2026c), gendered (Wikipedia, 2026a), and away from the Global South (Wikipedia, 2026b), and because GPTKB v1.5 finds ∼57% of materialized entities absent from it (Hu et al., 2025; Saeed et al., 2025b); a Wikipedia-only reference would mistake its own gaps for the model's.

Applied at scale, the measurement produces a consistent pattern. For `gpt-5-mini`, 1.2% of claims are refuted while 30.5% cannot be verified at all, and as expansion moves outward from the seed the true rate falls from 94.0% to 56.0% while precision barely moves and Wikipedia coverage drops to 51%. What deteriorates with distance is not the model's accuracy but the evidence available to judge it.

A natural objection is that grounding generation in retrieval would avoid this, and the most visible LLM-generated encyclopedia does exactly that without disclosing how (Grokipedia, 2025). We call the result the *retrieval trap*: lexical proximity to Wikipedia does not imply factual superiority, and may instead reflect rewriting that hides both what a model knows and where it fails. Testing the objection needs a system that retrieves nothing while generating (Figure 1, §5.4).

LLMpedia is that system, an open framework for parametric encyclopedia generation across `gpt-5-mini`, `Llama-3.3-70B`, and `DeepSeek-V3-0324`: ∼1M articles from GPT-5-mini and 120K each from the open-weight models in general-domain BFS, 27K in topic-focused runs over a controversy gradient (*Ancient Babylon*, *US Civil Rights*, *Dutch Colonization in Southeast Asia*) with two low-contestedness controls (*One Piece*, *Quantum Physics*), and 1K for the retrieval-trap benchmark.

Table 1 compares it with prior work on the three properties this argument needs at once: *scale*, since availability bias appears only past the documented core; *openness*, since a closed pipeline cannot be audited; and *verification granularity*, since an aggregate score cannot separate refuted from unverified. Three questions follow, each testing the claim under different conditions.

System	Param.	Fact. Full Art.	Serendip.	Scale
Benchmarks				
MMLU (Hendrycks et al., 2021)	✓	✗	✗	10^4
HLE (Phan et al., 2025)	✓	✗	✗	10^3
Knowledge materialization				
GPTKB (Hu et al., 2025)	✓	✗	✓	10^6
Cosmopedia (Ben Allal et al., 2024)	✓	✗	(✓)	10^7
Encyclopedia generation				
WikiSum (Liu et al., 2018)	✗	✓	✗	10^4
STORM (Shao et al., 2024)	✗	✓	✗	10^2
Grokipedia (Grokipedia, 2025)	✗(?)	✓	?	10^6
LLMpedia (ours)	✓	✓	✓	10^6

Table 1: Comparison across paradigms. Param.=parametric-only generation; Fact. Full Art.=factual full-article generation; Serendip.=not-preconceived knowledge discovery.

RQ1: What do topic-focused parametric encyclopedias reveal about cross-model factuality, coverage, and persona effects?

RQ2: At $120K$ scale, how do three model families differ in knowledge breadth, subject overlap, and factuality across BFS depth?

RQ3: Can parametric generation escape the Wikipedia retrieval trap - producing factually competitive but structurally independent content?

RQ1 holds domain and subject list fixed so the model is the only variable; RQ2 removes that constraint and follows the claim into the long tail; RQ3 rules out the alternative that a retrieval-grounded generator would not have the problem.

Our contributions are:

1. A fully open pipeline for parametric encyclopedia construction - prompts, artifacts, and verdicts.
2. Evidence that *availability bias is a first-order distortion*: true rate is 68.4% on Wikipedia-covered subjects (>21 pp below MMLU), driven by unverified claims (30.5%), not falsehood (1.2%); the judge is human-validated, and 90% of unverified verdicts audit to gaps in the reference sources.
3. Evidence that the beyond-Wikipedia frontier is real but difficult: 43% of subjects fall outside Wikipedia; among those with usable curated-web evidence, true rate is 57.6%.
4. A retrieval-trap benchmark where LLMpedia

outperforms Grokipedia at roughly half the Wikipedia similarity.

2 Related Work

Knowledge probing and its limits. LAMA (Petroni et al., 2019) probes recall via cloze prompts; MMLU (Hendrycks et al., 2021) spans knowledge-heavy domains; TruthfulQA (Lin et al., 2022) targets misconceptions; HLE (Phan et al., 2025) pushes frontier difficulty. All measure only what the experimenter thought to ask. LLMpedia complements benchmarks by materializing knowledge at scale and decomposing outputs into supported, refuted, and insufficient.

From triples to articles. Cohen et al. (2023) introduced recursive elicitation; Hu et al. (2025) scaled this into GPTKB (101M triples for 2.9M entities) and made availability bias the explicit target. Ghosh et al. (2025) mine the resulting KB for novel knowledge; Giordano and Razniewski (2026) find within-model variance from seed choice comparable to repeated-run variance, while cross-model variance is much larger. LLMpedia materializes discourse-level articles, enabling evaluation of structure, subject choice, and unverifiability in free text.

Atomic claim factuality. The decompose–retrieve–verify paradigm is standard for long-form factuality. FACTSCORE (Min et al., 2023) introduced it with binary *Supported/Not-supported* verification against Wikipedia; SAFE (Wei et al., 2024) extended retrieval to Google Search; VERISCORE (Song et al., 2024) added a three-label scheme restricted to verifiable claims; VERIFASTSCORE (Rajendhran et al., 2025) collapsed the pipeline but reverted to binary. We follow VeriScore's three labels, but spend additional effort on curating websites used for evidence retrieval, compared with generic web search. As evaluation costs are small in our case, we disregard local models like VeriScore and directly prompt commercial APIs.

Encyclopedia generation and the retrieval trap. STORM (Shao et al., 2024) models pre-writing through web-grounded interactions but remains retrieval-dependent. Gao et al. (2024) find outlines improve generation; LLMpedia extends this with dynamic, entity-tailored outlines. Grokipedia (Grokipedia, 2025) operates at the largest scale but discloses no methodology; Yasseri and Mohammadi (2025) report TF-IDF cosine of 0.46–0.49 against Wikipedia (LLMpedia: 0.256), suggesting retrieval-shaped repackaging.

Hallucination and provenance. Hallucination remains central to factual generation (Ji et al., 2023). RAG can improve factuality (Lewis et al., 2020; Mallen et al., 2023) but complicates provenance: once external passages are introduced, separating parametric memory from retrieved text becomes hard (Tuquero and PolitiFact, 2025; Yasseri and Mohammadi, 2025). LLMpedia avoids retrieval at generation time precisely to keep the two separable.

3 LLMpedia Framework

Our framework builds on the GPTKB methodology for LLM-based KB construction (Hu et al., 2025). Figure 3 shows the full pipeline. From a seed entity, the system expands breadth-first through optional self-grounding, outline generation, article elicitation, and three-stage entity sanitization before enqueuing surviving children. Prompts are in Appendix A.

3.1 Materialization Modes

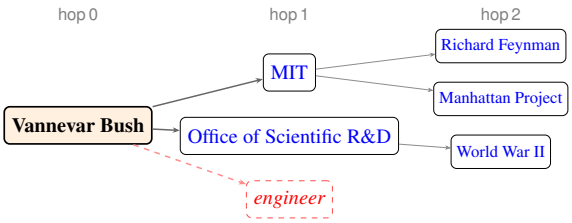


Figure 2: BFS from *Vannevar Bush*. Blue nodes survive sanitization; red dashed nodes are filtered as generic.

General-domain expansion. Starting from a single seed entity (*Vannevar Bush*, Figure 2), BFS expands without topical restrictions. The choice of seed is empirically inconsequential: Hu et al. (2025) construct a 100M-triple KB from the same seed, and Giordano and Razniewski (2026) systematically perturb seeds and find within-model variance comparable to repeated-run variance. Using BFS also matches prior materialization work (Cohen et al., 2023; Hu et al., 2025), enabling direct comparison. Each generated article is written in Wikitext with `[[wikilinks]]`; links that survive sanitization are enqueued as future article subjects. This mode scales to 10^5–10^6 articles.

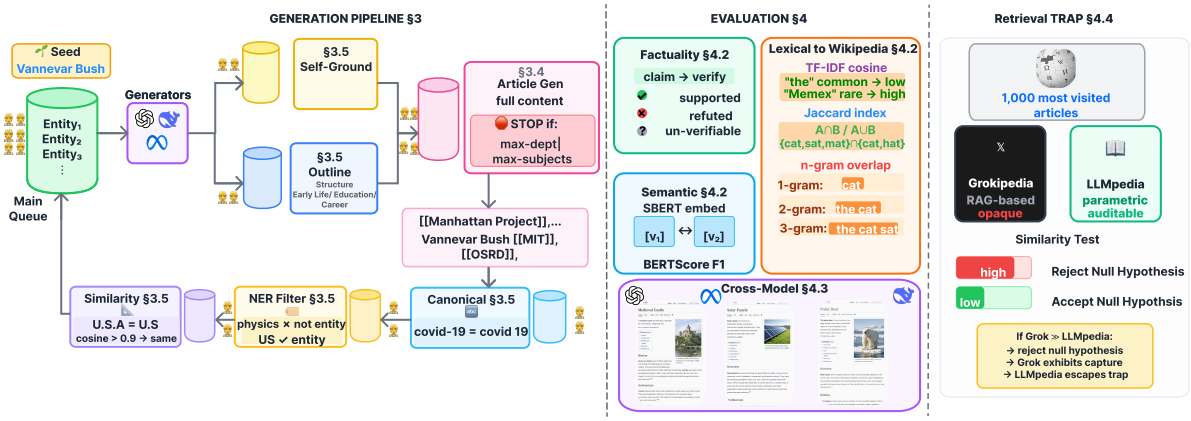


Figure 3: LLMpedia pipeline. Each subject flows through optional self-grounding, dynamic outline generation, and article elicitation. Extracted `[[wikilinks]]` undergo canonical normalization, LLM-based encyclopedic filtering, and embedding-based deduplication before surviving children enter the BFS queue. Details in Appendix B.

Topic-focused generation. Topic-focused BFS is constrained around a root topic (*Ancient Babylon*, *US Civil Rights Movement*, *Dutch Colonization in Southeast Asia*). Each run is seeded with a topic-specific subject list; generated `[[wikilinks]]` must remain topically associated (Appendix A); only relevant survivors are re-enqueued. This yields a controlled regime for cross-model, persona, and domain-difficulty analysis.

(a) Outline *(three sections, adapted to entity)*

```
Vannevar Bush
1. Early life and education
2. Engineering career and Raytheon
3. OSRD and wartime science leadership
```

(b) Generated Wikitext

```
'''Vannevar Bush''' was an American
[[engineer]] who headed the [[Office
of Scientific R&D]] during [[World War
II]]...
== Early life and education ==
Bush was born in [[Everett,
Massachusetts]] and studied at [[Tufts
University]]...
== OSRD and wartime science leadership
==
In 1940, Bush proposed the formation
of the [[NDRC]]...
```

Figure 4: The model proposes an entity-tailored outline (top, blue) then writes the article (bottom, orange) under verbatim section headers. Blue wikilinks survive sanitization; red `[[engineer]]` is filtered as generic.

3.2 Outline, Generation, and Prompting

Outlines improve LLM-based encyclopedia generation (Gao et al., 2024), typically with fixed templates. LLMpedia proposes 6–7 section headings tailored to the entity, then writes Wikitext to match (Figure 4).

Optional self-grounding (He and Razniewski, 2025) produces a per-entity fact sheet (summary, aliases, predicate–object pairs with confidence $\geq$ 0.75); we treat it as an ablation variable (§4.4). The *baseline* prompt requests plain wikilinks; the *calibrated* prompt requires per-entity confidence (`[[Entity (0.85)]]`, range 0.75–1.00) inspired by (Tian et al., 2023); entities below 0.75 are discarded before NER.

3.3 Entity Sanitization Pipeline

GPTKB v1 and v1.5 document persistent entity disambiguation failures (Hu et al., 2025). We address these with three stages, each with a distinct role. *Stage* refers to these three throughout; the generation steps (outline, elicitation, wikilink proposal, NER) are not stages.

Stage 1: Canonical Normalization (re-encounter dedup). Unicode NFKC, case-folding, and punctuation removal map surface variants of the *same* entity to identical keys ("U.S."/"US"→`us`; "Covid-19"/"Covid 19"→`covid19`). This collapses re-encounters of already-committed entities; *it is not entity loss*. The large raw-to-canon reduction (70.2M→2.29M for GPT-5-mini) reflects natural re-linking - every article re-mentions common entities the pipeline has already canonicalized. Stage 1 cannot disambiguate homonyms (e.g., *Apple* the company vs. the fruit); that is Stage 3's job.

Stage 2: LLM-Based Encyclopedic Filtering. Not every wikilink warrants its own article. Generic forms like `[[engineer]]` (Figure 2) produce thin, looping sub-articles; in contrast, `[[Office of Scientific Research and Development]]` expands into a self-contained article. Following Hu et al. (2025) we use the LLM itself to classify candidate batches. The filter is intentionally conservative: a candidate must be positively classified non-encyclopedic to drop, and filtered candidates can be re-nominated at later BFS waves.

Stage 3: Semantic Dedup and Sense Disambiguation. This stage handles two distinct cases: (i) semantic duplicates with different surface forms (*Deutschland*/*Germany*, *Oxford University*/*University of Oxford*), and (ii) true homonyms with identical surface forms (*Apple* the company vs. the fruit). We maintain an embedding index (`text-embedding-3-small`) over committed entities; candidates above cosine 0.90 trigger LLM arbitration over the first ∼30 words of each candidate's lead paragraph plus parent context. For case (i), the leads converge and the candidates merge; for case (ii), the leads diverge semantically (Apple Inc. discusses Cupertino, consumer electronics; Apple the fruit discusses *Malus domestica*, cultivars) and arbitration keeps them separate. Within-wave deduplication merges concurrent proposals before commit (Appendix B.2). Homonyms with identical surface forms *and* closely related leads (e.g., *Dresden* the German city vs. U.S. locality) remain a known limitation (§7).

Architecture and persona injection. Each stage has an independent work queue with retries and exponential backoff, supporting batch and online modes (Appendix B.1). To study framing as an explicit variable, we define three personas (scientific-neutral, left-leaning, conservative) as fixed instruction strings and prepend the active one to the *system message* of every call in a topic-focused run: outline, elicitation, wikilink proposal, and entity filtering. "System level" means the system role of the chat template, not a separate processing step; the three strings are given verbatim in Appendix A.

4 Experimental Setup

4.1 Models and Conditions

We evaluate `gpt-5-mini` (OpenAI, 2025a), `Llama-3.3-70B-Instruct` (Grattafiori et al., 2024) (4×A100 on-prem), and `DeepSeek-V3-0324` (671B/37B MoE, 16×H100 on-prem) (Zhao et al., 2025). All runs use temperature 0 with fixed seeds. GPT-5-mini is primary for instruction-following compliance and cost efficiency; the open-weight models test generalization beyond a single API.

Topic-focused (RQ1). Motivated by bias concerns around Grokipedia (Grokipedia, 2025; Yasseri and Mohammadi, 2025), we test whether persona injection produces measurable framing shifts on the same subject. We span a controversy gradient: *US Civil Rights* and *Dutch Colonization in Southeast Asia* as contested narratives, *Ancient Babylon* as a politically neutral control. Each topic seeds ∼1K subjects under three personas across all three models - 27 cells, ∼27K articles.

Large-scale materialization (RQ2). All three models start from *Vannevar Bush*. GPT-5-mini produces ∼1M articles; open-weight models produce ∼120K each. We compare on the first 120K subjects.

Retrieval-trap (RQ3). Following Yasseri and Mohammadi (2025), we collect the 1,000 most-edited English Wikipedia articles confirmed present on Grokipedia and evaluate both systems against Wikipedia (Tier 1) and curated web sources (Tier 2).

4.2 Evaluation Protocol

We adopt the decompose–retrieve–verify paradigm established by FACTSCORE (Min et al., 2023) and extended by SAFE (Wei et al., 2024), VERISCORE (Song et al., 2024), and VERIFASTSCORE (Rajendhran et al., 2025): decompose each article into atomic claims, retrieve evidence per subject, and verify each claim independently against the retrieved evidence. We follow VERISCORE in using a three-label verdict scheme (*supported* / *refuted* / *insufficient*) and pair it with a two-tier evidence stack: Wikipedia, and high-quality web sources reaching subjects Wikipedia does not cover. The judge is `gpt-4.1-nano`; we validate it directly against human verdicts in §4.3.

Choice of judge. Three design choices justify our verification pipeline. *First, the label scheme.* VeriScore's three labels (vs. binary) prevent collapsing contradicted and evidence-silent claims into a single bucket - the distinction that makes the 1.2% false vs. 30.5% unverified gap, the signature of availability bias, measurable. Under a binary scheme, absence of evidence reads as not-supported; our prompt states the asymmetry as a hard rule with worked examples, so silence yields *insufficient* and never *refuted*, and a parser check re-labels the 2.3% of refuted verdicts whose explanation cites silence (Appendix C). *Second, the judge is human-validated.* Following FActScore's protocol, we audit `gpt-4.1-nano` on balanced 33/33/33 samples from both tiers (§4.3): per-class agreement on *supported* and *insufficient* reaches **87.9–93.9%**, with disagreements cancelling in aggregate. A FActScore cross-check on a sample returned 89.8% supported, broadly consistent though only partially comparable across label schemes. *Third, the evidence.* Rather than VeriScore's open Google Search or FActScore's Wikipedia-only retrieval, we admit sources by quality rule rather than by open search (Appendix D) - reaching the 43% of subjects Wikipedia omits while excluding the long tail of low-quality pages an unconstrained Google search would surface.

Stage 1 - Claim extraction. The article is sent to the judge with an extraction prompt asking for up to 10 atomic, self-contained claims; each must mention the subject explicitly, state one predicate, and avoid opinion or hedging. Verifiability constraints mirror VERISCORE: events and states with necessary modifiers, excluding opinions, hypotheticals, and instructions.

Stage 2 - Evidence retrieval. Evidence is retrieved *independently* of the generated article, using only the subject name as query, so the article cannot shape its own evidence. **Tier 1 (Wikipedia)** fetches the full article via MediaWiki with redirect resolution; subjects Wikipedia does not cover contribute to coverage but not Tier 1 factuality. **Tier 2 (curated web)** is governed by a source-quality rule, not by a list. Each candidate URL is scored 0–100 and fetched only above 60. Any institutional or editorially accountable source qualifies: every `.gov`, `.edu`, `.com` and `.org` domain is admitted at a default score of 70 whether or not we have seen it before, so the set of admissible sources is open-ended and much larger than anything we could enumerate. The ∼133 domains in Appendix D are hand-scored examples that calibrate the scale - encyclopedias, national archives, wire services, academic publishers - not the sources the pipeline is limited to. Commercial and unattributed sites default to 35 and are excluded, as are Wikipedia and its mirrors. Up to three admissible URLs per subject are fetched in descending score order and concatenated.

On long-form vs. targeted elicitation. Long-form generation entangles parametric knowledge retrieval with composition, and we do not claim to fully separate them. Decomposition with independent per-claim verification factors out composition at the verification step, with the prompt restricted to propositional support. Targeted elicitation (Petroni et al., 2019; Sun et al., 2024) measures knowledge in isolation but sacrifices the modality users consume - our long-form regime is complementary.

Metrics. With N total claims, n_s supported, n_r refuted and n_u insufficient, $\text{Prec} = n_s/(n_s{+}n_r)$, $\text{True} = n_s/N$, $\text{False} = n_r/N$ and $\text{Unv} = n_u/N$. Precision excludes unverified claims; *True*, *False* and *Unv* sum to 1. *Unv* means the retrieved evidence does not settle the claim, not that the claim is false. All metrics are macro-averaged at article level.

Similarity. TF-IDF cosine, token Jaccard, n-gram overlap ($n{=}1{-}3$), and semantic cosine (`text-embedding-3-small`) between generated articles and Wikipedia.

4.3 Human Validation

To audit the LLM judge, we drew balanced stratified samples from its verdict distribution and asked a human annotator, one of the authors, working from the same evidence snippets as the judge, to go through them to agree/disagree and supply the correct class otherwise. The instructions restate the three definitions given to the judge (Appendix C), including the rule that silence yields *insufficient* and never *refuted*. TIER 1 (WIKIPEDIA): 99 (claim, evidence, verdict) triples balanced 33/33/33; TIER 2 (CURATED WEB): 30 triples balanced 10/10/10 from the frontier-Web subset. Agreement is the share of a stratum where the annotator endorsed the judge; on the contribution-bearing classes (*sup-*

ported, driving true rate; *insufficient*, driving the unverified rate) it reaches **87.9–93.9%** across both tiers. Per-claim disagreements largely cancel in aggregate: human totals are 37/24/38 (Tier 1) and 13/7/10 (Tier 2), so *supported* drifts by only +4 of 33 and +3 of 10, keeping the macro estimate robust. Full details are in Appendix E. The verdict set these samples are drawn from - 2,010 subjects and 20,092 claims - is released with the evidence behind each verdict at `https://llmpedia.net`, so the judge's output can be re-audited independently.

4.4 Ablation

A 2×2×2 factorial ablation over **self-grounding** (SG; on/off), **prompting strategy** (baseline vs. calibrated), and **reasoning budget** (minimal vs. low) on `gpt-5-mini` (500 articles, seed 42); full design in Appendix F. Self-grounding with calibration costs ∼33% more for only +1.6 pp precision over the operating point. We adopt **no SG + baseline + minimal**† for all large-scale runs, prioritizing reproducibility and cost.

SG	Prom.	Reas.	Prec	True	False	Unv
✓	calib	min	**96.3**	**93.7**	3.4	2.9
✓	base	min	96.2	93.0	3.6	3.4
✗	calib	min	94.5	89.7	4.5	5.8
✗	base	min	94.7†	89.5	4.5	6.0

Table 2: Ablation on `gpt-5-mini` (minimal reasoning, 500 articles, %). SG = self-grounding. † = operating point.

5 Results

5.1 Pipeline Validation: Sanitization Funnel

Every wikilink passes three filters before becoming an article subject: canonical normalization, NER filtering, and semantic deduplication (§3.3). They discard the large majority of candidates, so this subsection reports what they discard: the RQ results hold only if the funnel removes redundancy and not knowledge.

GPT-5-mini produces 1,008,947 articles and 70,165,356 raw wikilink mentions, consolidated to 12,479,792 pre-NER candidates. Canonical deduplication keeps 2,293,089 (18.4%); NER accepts 1,654,284; Stage 3 accepts 1,120,843; 1,063,929 children enter the BFS queue - 1.52% of raw mentions (Table 3). DeepSeek and Llama show the same funnel shape at 120K-article scale

	GPT-5-mini	DeepSeek	Llama
Generated articles	1,008,947	120,139	120,100
Raw `[[links]]`	70,165,356	6,490,752	9,820,122
Pre-NER input	12,479,792	1,999,933	4,326,286
After canonical dedup	2,293,089	858,516	1,247,222
After NER	1,654,284	443,736	683,681
After similarity	1,120,843	396,254	499,245
Queue-inserted children	1,063,929	396,090	498,225
Raw → queue	1.52%	6.10%	5.07%
Entities w/ aliases	108,030	14,591	26,258
Aliases absorbed	527,469	35,558	101,548
Aliases per entity	4.88	2.44	3.87

Table 3: Cross-model sanitization funnel and alias absorption. Aliases per entity is averaged over entities that absorbed at least one variant. Full per-stage survival in Appendix G.

with higher raw-to-queue survival (6.10%, 5.07%) because their committed indexes are smaller (Appendix G).

Alias absorption shows that Stage 3 collapses variants rather than dropping entities. Each rejection records the canonical entity the candidate duplicated, so these counts are the near-duplicate articles the pipeline would otherwise have written: 108,030 GPT-5-mini entities absorbed 527,469 variants, 4.88 each. The heaviest absorbers are U.S. federal departments, universities under acronym and *University of X* / *X University* alternations, places with and without state qualifiers, and organizations under name and acronym (*University of Oxford* alone absorbed 1,814) - the variant spaces canonical keying cannot reach. Under 2% of a sampled set are arguable, and none affect the factuality numbers (Appendix G.6).

5.2 RQ1: Topic-Focused Analysis

GPT-5-mini leads precision across all three topics (98.3–99.4%), followed by DeepSeek (95.9–97.3%) and Llama-70B (93.7–96.5%), with false-claim rates compressed below 3% everywhere (Table 4). The dominant cross-model signal is *unverifiability*: Llama exceeds GPT by 11.3 pp on US Civil Rights, 9.6 pp on Ancient Babylon, and 12.0 pp on Dutch Colonization - the same availability-bias pattern that emerges at scale (§5.3). Dutch Colonization is hardest (Llama 35.1% unverifiable), reflecting weaker Wikipedia coverage of the non-Anglophone domain (W% 60.7). Cross-model canonical entity overlap is low (CaJ 0.17–0.22): even on the same topic, models

foreground substantially different entities, consistent with Giordano and Razniewski (2026).

Topic	Model	Prec	Hall	Unv	W%	CaJ
Ancient Babylon	GPT-5-mini	**99.4**	0.5	18.8	71.0	
	DeepSeek	97.3	1.8	24.6	75.7	.19
	Llama-70B	93.7	2.9	28.4	79.7	
US Civil Rights	GPT-5-mini	**98.3**	1.5	10.8	78.3	
	DeepSeek	96.9	2.3	17.4	83.0	.22
	Llama-70B	96.5	2.4	22.1	75.3	
Dutch Colon.	GPT-5-mini	**98.8**	0.8	23.1	66.3	
	DeepSeek	95.9	2.3	24.8	70.3	.17
	Llama-70B	95.2	2.5	35.1	60.7	

Table 4: Topic-focused results (persona-averaged). **Prec**: precision; **Hall**: false rate; **Unv**: unverifiable rate; **W%**: Wikipedia coverage; **CaJ**: avg. canonical entity Jaccard across model pairs. Factuality on Wikipedia-covered subjects.

Persona effects. Each article is scored on a 24-dimensional topic-relevant lexicon (Appendix H) per 1,000 content tokens using a deterministic word-list classifier. On intersection-sampled subjects (30 per model×persona×topic cell), paired Wilcoxon signed-rank tests with Bonferroni correction over 648 comparisons yield 37 significant effects in topically expected directions. On Dutch Colonization, left-leaning uses colonized-side vocabulary (*exploitation*, *plunder*, *dispossession*) 5.6 hits per 1,000 tokens more than conservative; conservative flips toward development framing (+2.3). On US Civil Rights, left-leaning skews progressive (+4.6 vs. conservative) and foregrounds grassroots organizers (+1.8); conservative centres canonical institutions. On Ancient Babylon - the neutral control - contested political axes collapse to non-significance. Factual precision is unchanged (≤3.6 pp between any two personas in the same cell): persona changes *what* the article says, not *how often it is right*. Extending the analysis to two non-contested domains (*Quantum Physics*, *One Piece*; Appendix H.6) yields only **6** Bonferroni-significant shifts versus 37 on the primary topics, confirming that persona framing is triggered by domain-specific ideological affordances, not applied indiscriminately.

5.3 RQ2: Cross-Model at 120K

All three models start from the same seed under general-domain BFS. Despite the shared seed, only 7.3% of subjects appear in all three corpora (Table 5); early wikilink choices cascade into substantially different graphs. The robustness analysis of Giordano and Razniewski (2026) shows within-model variance matches repeated-run variance, ruling out BFS noise - the 7.3% overlap reflects genuine cross-model knowledge divergence.

A consistent precision-vs-true-rate gap emerges. Precision is preserved from shared to independent (max drop 1.1 pp); true rate falls sharply (GPT 88.6→78.6; DeepSeek 85.8→77.6; Llama 79.1→64.5; max drop 14.6 pp), tracking Wikipedia coverage (94.0%→70.6–72.4%). The shared intersection is the canonical core all three models agree is encyclopedic; once we leave it for each model's long tail, models remain right about what they assert but external evidence increasingly cannot verify it. The ranking GPT-5-mini > DeepSeek > Llama is stable across conditions.

	GPT-5-mini	DeepSeek	Llama-70B
Corpus (120K each)			
Mean `[[links]]`/art	70.6	51.5	81.0
Mean sections	6.0	5.2	2.1
Mean words	885	739	845
Subject overlap (120K×3)			
Union / Intersection	280,494 / 20,417 (7.3%)		
Shared intersection (1K, Tier 1)			
Wiki coverage	94.0%	94.0%	94.0%
Precision	**98.5**	96.1	95.2
True rate	**88.6**	85.8	79.1
Independent (1K, Tier 1)			
Wiki coverage	72.4%	70.6%	72.4%
Precision	**97.8**	95.5	94.1
True rate	**78.6**	77.6	64.5

Table 5: Cross-model comparison at 120K. Factuality on Wikipedia-covered subjects only. Supplementary figures in Appendix M.

Hop-Stratified and Frontier Results

Table 6 is the empirical backbone of the paper. On a uniform-random 1,000-article sample from the ∼1M GPT-5-mini corpus, Wikipedia covers only 567 subjects (56.7%). On those, true rate is 68.4%, false 1.2%, unverifiable 30.5%. The picture sharpens with BFS depth: true rate falls from 94.0% at hop 1 to 56.0% at hop 6 - a 38-point degradation. This collapse is not driven by hallucination: false rate stays <2% at every depth and precision moves marginally (97.9→96.5%), similar to Hu et al. (2025).

What grows in lockstep with the true-rate drop is the *unverifiable* rate (4%→43%), tracking the parallel collapse in Wikipedia coverage (100%→51%). The model is not asserting more

Bucket	Ref	Sampled	Found	Cov %	Prec	True	False	Unv
hop 0	Wiki	1	1	100	100.0	**70.0**	0.0	30.0
hop 1	Wiki	10	10	100	97.9	**94.0**	2.0	4.0
hop 2	Wiki	200	184	92	98.7	**84.6**	1.1	14.4
hop 3	Wiki	200	168	84	98.7	**80.3**	0.6	19.1
hop 4	Wiki	200	134	67	97.7	**72.6**	1.4	26.0
hop 5	Wiki	200	111	55.5	97.6	**69.0**	1.2	29.8
hop 6	Wiki	200	102	51	96.5	**56.0**	1.0	43.0
random	Wiki	1000	567	56.7	97.1	68.4	1.2	30.5
random	Web	1000	779	77.9	97.7	56.5	0.7	42.8
frontier	Web	433	311	71.8	98.3	57.6	0.6	41.8

Table 6: GPT-5-mini factuality by BFS depth and evidence tier. **Sampled**: subjects attempted; **Found**: with usable evidence; **Cov**: Found/Sampled. Frontier row restricts to subjects absent from Wikipedia. All rates %.

falsehoods at depth; evaluation cannot reach the knowledge it does encode. This is the signature availability bias predicts: fixed-question benchmarks miss the long tail because the long tail is, by construction, not in the benchmark. On the web tier, 77.9% of random-sample subjects yield usable evidence; among 433 frontier subjects absent from Wikipedia, 311 (71.8%) yield usable evidence with true rate 57.6% and precision 98.3% - the benchmark-vs-open-ended gap is a coverage problem, not a hallucination problem.

What unverified verdicts are. Retrieval failure is excluded by construction: rates are computed only over subjects with usable evidence (**Found** in Table 6), and the remainder are reported as coverage, so an unverified verdict means evidence was collected and the claim is absent from it. To separate that from evaluator error, the annotator of §4.3 audited 100 stratified unverified verdicts from Wikipedia-covered subjects, reading each claim beside the judge's evidence and assigning a cause (Table 7). Evidence silence accounts for 90%, with no case from entity ambiguity or claim extraction, and correcting the three should-be-refuted cases moves the false rate by under half a point. Silence also stays at 78.6% or above from hop 4 on (Appendix I), where evaluator error would have no reason to grow. A residual share can still come from evaluator-side causes, so we read the unverified rate as an upper bound on the availability gap.

5.4 RQ3: Retrieval Trap

Table 8 shows the central pattern. LLMpedia leads Grokipedia by **+6.9 pp** true rate against Wikipedia and **+2.9 pp** on the web tier. Grokipedia's false rate (1.8%) is more than 2× LLMpedia's (0.8%) under Wikipedia verification.

Cause of unverified verdict	n	% (95% CI)
Evidence silence	90	90.0 (82.6–94.5)
Entity-ambiguity error	0	0.0 (0.0–3.7)
Claim-extraction defect	0	0.0 (0.0–3.7)
Judge error, should be supported	7	7.0 (3.4–13.7)
Judge error, should be refuted	3	3.0 (1.0–8.5)

Table 7: Causes of 100 audited unverified verdicts, Wilson 95% CIs. Breakdown by hop in Appendix I.

Metric	LLMpedia	Grokipedia
Factuality (Wiki / Web)		
Precision (%)	**98.6 / 98.8**	97.3 / 97.0
True rate (%)	**86.0 / 76.3**	79.1 / 73.4
False rate (%)	**0.8 / 0.7**	1.8 / 1.7
Unverifiable (%)	13.3 / 23.0	19.1 / 24.9
Similarity to Wikipedia		
TF-IDF cosine	0.256	0.493
Bigram overlap	0.090	0.200
Trigram overlap	0.026	0.079
Semantic cosine	0.773	0.811
Words (mean / med.)	2,016 / 1,958	7,376 / 6,193

Table 8: Retrieval-trap on 1,000 most-edited Wikipedia articles. Factuality: **Tier 1 (Wiki) / Tier 2 (Web)**. Full breakdown with standard deviations in Appendix J.

LLMpedia's unverifiable rate is also lower (13.3% vs. 19.1% Wiki). LLMpedia achieves this at roughly half the TF-IDF cosine (0.256 vs. 0.493) and much lower n-gram overlap, while semantic cosine remains comparable because both systems discuss the same subjects. This is the retrieval trap: Grokipedia stays close to Wikipedia's surface form yet is less factually reliable - consistent with retrieval-shaped rewriting without robust entity disambiguation (Appendix K). Lexical proximity to Wikipedia is not a proxy for truth.

6 Conclusion

LLMpedia materializes ∼1.3M articles from parametric memory across three model families, audited against Wikipedia and curated web evidence. `gpt-5-mini` reaches 68.4% true rate, 21 pp below MMLU, driven by unverified claims (30.5%) not falsehood (1.2%), 90% of which audit to evidence gaps. Beyond Wikipedia the frontier is hard: 43% of subjects lie outside it, 57.6% verified on the web tier. Lexical proximity is no proxy for truth: LLMpedia beats Grokipedia at half its TF-IDF similarity. `https://llmpedia.net`.

7 Limitations

Single-pass sampling and capability conflation. Every article is one generation under fixed prompt and temperature-0 decoding, surfacing a sample of latent knowledge rather than the entirety. Self-consistency (Wang et al., 2023) could recover more but multiplies the dominant generation cost (already ∼$3,500; Appendix L). Long-form prose also entangles latent knowledge retrieval with composition under context; decomposition mitigates this at the verification step but not the generation step. Targeted single-claim elicitation (Petroni et al., 2019; Sun et al., 2024) measures knowledge in isolation but loses the modality users consume; pairing both protocols on the same entity is a natural extension.

Verifier choice. We follow the decompose–retrieve–verify pipeline of FActScore and VeriScore but use a general-purpose judge (`gpt-4.1-nano`) rather than a fine-tuned verifier. Their verifiers were developed and validated at ∼6.5K-generation scale; we validate the judge directly against human verdicts on balanced 33/33/33 samples (§4.3) and invest the additional rigor on the evidence side (Appendix D). A FActScore cross-check on sample returned 89.8% factuality, broadly consistent with our judge's scores, though direct comparison is partial because of the binary-vs-three-label difference. The audit in §5.3 puts evaluator-side error in the unverified class near 10% on 100 samples, so the reported unverified rate is an upper bound on the availability gap; both exercises used one expert annotator, so we report agreement with the judge rather than inter-annotator agreement.

What the source rule admits. Tier 2 admits sources by quality score, and any such rule encodes a view of what counts as a reference. Institutional sources we never enumerated enter through the `.gov`/`.edu`/`.org` default, but sources that are authoritative in a domain without carrying those signals - local archives, non-Anglophone outlets, community-maintained references - fall below the bar and are never fetched.

Temporal snapshot and selection bias. All runs were conducted between January and March 2026; Wikipedia and web references evolve continuously, so a claim judged unverified today may become verifiable or refutable later. Separately, 28.2% of frontier subjects yield no usable web evidence under strict source-quality filtering, potentially biasing Tier 2 estimates toward better-documented subjects.

Scale asymmetry. Only `gpt-5-mini` reaches ∼1M articles; the open-weight models are constrained to ∼120K each, so the deepest hop-stratified analysis is GPT-5-mini specific.

Scope of measurement. Our framework captures propositional factual support, not coherence, neutrality, salience, completeness, or omission. An article can achieve high precision while being editorially unbalanced. Our goal is to measure *how much knowledge models surface* in the long-form modality users consume, not to certify articles as well-rounded encyclopedic reference; the latter requires an orthogonal evaluation we leave to future work.

Homonymy and privacy. Stage 3 LLM arbitration handles two regimes well: it merges semantic duplicates with different surface forms (*Deutschland*/*Germany*, *U.S.*/*USA*) by detecting that their lead paragraphs converge, and it keeps *qualified* homonyms (*Apple Inc.* vs. *Apple*; *Dresden, Germany* vs. *Dresden, Maine*) separate because their distinct surface forms yield distinct canonical keys and distinct embeddings before arbitration even runs. The failure mode is *bare* homonymy: two genuinely distinct entities surfaced under identical, unqualified strings (e.g., both written simply as *Dresden* - the German city and a U.S. locality). In that case the surface form is the only key the canonical-dedup stage sees, and the second arrival is treated as a re-encounter of the first; the two senses are merged before Stage 3's lead-based arbitration is ever consulted, even though their leads would in principle separate them. Resolving this requires context-aware entity linking with sense-specific identifiers, which we leave to future work. Separately, generated articles may mention real public figures because encyclopedic subjects include them; the pipeline is not designed to expose private information, and any inadvertent occurrence is filtered from releases.

References

Anthropic. 2026. The anthropic economic index understanding ai's effects on the economy. https://www.anthropic.com/economic-index. Accessed May 2026.

Loubna Ben Allal, Anton Lozhkov, and Daniel van Strien. 2024. Cosmopedia: how to create large-scale synthetic data for pre-training. Hugging Face Blog.

Aaron Chatterji, Thomas Cunningham, David J Deming, Zoe Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman. 2025. How people use chatgpt. Technical report, National Bureau of Economic Research.

Roi Cohen, Mor Geva, Jonathan Berant, and Amir Globerson. 2023. Crawling the internal knowledge-base of language models. In *Findings of the Association for Computational Linguistics: EACL 2023*, pages 1856–1869, Dubrovnik, Croatia. Association for Computational Linguistics.

Fan Gao, Hang Jiang, Rui Yang, Qingcheng Zeng, Jinghui Lu, Moritz Blum, Tianwei She, Yuang Jiang, and Irene Li. 2024. Evaluating large language models on Wikipedia-style survey generation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 5405–5418, Bangkok, Thailand. Association for Computational Linguistics.

Shrestha Ghosh, Luca Giordano, Yujia Hu, Tuan-Phong Nguyen, and Simon Razniewski. 2025. Mining the mind: What 100m beliefs reveal about frontier llm knowledge. *arXiv preprint arXiv:2510.07024*.

Luca Giordano and Simon Razniewski. 2026. Foundations of LLM knowledge materialization: Termination, reproducibility, robustness. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 2145–2164, Rabat, Morocco. Association for Computational Linguistics.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, and 1 others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Grokipedia. 2025. Grokipedia. https://grokipedia.com/.

Kunal Handa, Alex Tamkin, Miles McCain, Saffron Huang, Esin Durmus, Sarah Heck, Jared Mueller, Jerry Hong, Stuart Ritchie, Tim Belonax, and 1 others. 2025. Which economic tasks are performed with ai? evidence from millions of claude conversations. *arXiv preprint arXiv:2503.04761*.

Jingbo He and Simon Razniewski. 2025. Combining self-retrieval-augmented generation with divide-and-conquer for language model-based knowledge base construction. In *KBC-LM/LM-KBC@ ISWC*.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. In *ICLR*. OpenReview.net.

Yujia Hu, Tuan-Phong Nguyen, Shrestha Ghosh, and Simon Razniewski. 2025. Enabling LLM knowledge analysis via extensive materialization. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16189–16202, Vienna, Austria. Association for Computational Linguistics.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Comput. Surv.*, 55(12).

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS '20, Red Hook, NY, USA. Curran Associates Inc.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.

Peter J. Liu, Mohammad Saleh, Etienne Pot, Ben Goodrich, Ryan Sepassi, Lukasz Kaiser, and Noam Shazeer. 2018. Generating wikipedia by summarizing long sequences. *arXiv:1801.10198 [cs]*.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822, Toronto, Canada. Association for Computational Linguistics.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100.

OpenAI. 2025a. GPT-5-mini. Accessed: 2026-03-17.

OpenAI. 2025b. Wie menschen chatgpt nutzen.

Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. 2019. Language models as knowledge bases? In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2463–2473, Hong Kong, China. Association for Computational Linguistics.

Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, Chen Bo Calvin Zhang, Mohamed Shaaban, John Ling, Sean Shi, and 1 others. 2025. Humanity's last exam. *arXiv preprint arXiv:2501.14249*.

Rishanth Rajendhran, Amir Zadeh, Matthew Sarte, Chuan Li, and Mohit Iyyer. 2025. VeriFastScore: Speeding up long-form factuality evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 9234–9259, Suzhou, China. Association for Computational Linguistics.

Muhammed Saeed, Muhammad Abdul-Mageed, and Shady Shehata. 2025a. Surfacing subtle stereotypes: A multilingual, debate-oriented evaluation of modern llms. *arXiv preprint arXiv:2511.01187*.

Muhammed Saeed, Shaina Raza, Ashmal Vayani, Muhammad Abdul-Mageed, Ali Emami, and Shady Shehata. 2025b. Beyond content: How grammatical gender shapes visual representation in text-to-image models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 24673–24695, Suzhou, China. Association for Computational Linguistics.

Yijia Shao, Yucheng Jiang, Theodore Kanell, Peter Xu, Omar Khattab, and Monica Lam. 2024. Assisting in writing Wikipedia-like articles from scratch with large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6252–6278, Mexico City, Mexico. Association for Computational Linguistics.

Yixiao Song, Yekyung Kim, and Mohit Iyyer. 2024. VeriScore: Evaluating the factuality of verifiable claims in long-form text generation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 9447–9474, Miami, Florida, USA. Association for Computational Linguistics.

Kai Sun, Yifan Xu, Hanwen Zha, Yue Liu, and Xin Luna Dong. 2024. Head-to-tail: How knowledgeable are large language models (LLMs)? A.K.A. will LLMs replace knowledge graphs? In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 311–325, Mexico City, Mexico. Association for Computational Linguistics.

Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher Manning. 2023. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5433–5442, Singapore. Association for Computational Linguistics.

Loreben Tuquero and PolitiFact. 2025. What's Grokipedia, Musk's AI-powered rival to Wikipedia? Al Jazeera.

Amos Tversky and Daniel Kahneman. 1973. Availability: A heuristic for judging frequency and probability. *Cognitive psychology*, 5(2):207–232.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.

Jerry Wei, Chengrun Yang, Xinying Song, Yifeng Lu, Nathan Hu, Jie Huang, Dustin Tran, Daiyi Peng, Ruibo Liu, Da Huang, Cosmo Du, and Quoc V. Le. 2024. Long-form factuality in large language models. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, NIPS '24, Red Hook, NY, USA. Curran Associates Inc.

Wikipedia. 2026a. Gender bias on Wikipedia. https://en.wikipedia.org/wiki/Gender_bias_on_Wikipedia.

Wikipedia. 2026b. Wikipedia:Size of Wikipedia. https://en.wikipedia.org/wiki/Wikipedia:Size_of_Wikipedia.

Wikipedia. 2026c. Wikipedia:Systemic bias. https://en.wikipedia.org/wiki/Wikipedia:Systemic_bias.

Taha Yasseri and Saeedeh Mohammadi. 2025. How similar are grokipedia and wikipedia? a multidimensional textual and structural comparison. *arXiv preprint arXiv:2510.26899*.

Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Huazuo Gao, Jiashi Li, Liyue Zhang, Panpan Huang, Shangyan Zhou, Shirong Ma, Wenfeng Liang, Ying He, Yuqing Wang, Yuxuan Liu, and Y.X. Wei. 2025. Insights into deepseek-v3: Scaling challenges and reflections on hardware for ai architectures. In *Proceedings of the 52nd Annual International Symposium on Computer Architecture*, ISCA '25, page 1731–1745, New York, NY, USA. Association for Computing Machinery.

A Prompt Templates

LLMPedia operates under two **generation regimes** and two **elicitation strategies**, yielding four prompt configurations. The **general-domain** regime performs open BFS expansion from an arbitrary seed with no topical constraint. The **topic-focused** regime also performs BFS but restricts expansion to entities with a direct, meaningful connection to a designated root topic (e.g., *Ancient Babylon*). Orthogonally,

baseline prompts request standard Wikitext with plain `[[wikilinks]]`, while **calibrated** prompts additionally require each wikilink to be annotated with a confidence score (`[[Entity (0.85)]]`) reflecting the model's certainty that the entity belongs to the subject's semantic neighborhood; links below threshold $\tau = 0.75$ are discarded by the NER stage before reaching similarity arbitration.

Each subject passes through two prompt roles in sequence:

1. **Article elicitation**: generates the full Wikitext article following a pre-generated section outline. The calibrated variant requires scored wikilinks.
2. **NER filtering**: classifies each raw wikilink extracted from the article as a true named entity worthy of BFS expansion or a reject. The calibrated variant additionally assigns confidence scores.

Runtime placeholders. All prompts share the following runtime-filled placeholders:

- {`subject_name`}: the entity currently being processed, e.g. *Vannevar Bush*.
- {`root_subject`}: the root topic constraining topic-focused runs, e.g. *Ancient Babylon*; absent in general-domain prompts.
- {`avg_words_per_article`}: target article length in words (default 716, matching Wikipedia's overall average (Wikipedia, 2026b)).
- {`outline_block`}: the JSON section list produced by the outline generation step and injected verbatim into elicitation prompts; see Example A below.
- {`phrases_block`}: the newline-separated list of raw wikilink candidates extracted from the generated article, passed to NER filtering; see Example A below.
- {`persona_block`}: one of three fixed instruction strings injected at the system level of *every* pipeline stage-elicitation, NER, and similarity arbitration-to probe how editorial stance shapes content selection and framing independently of subject matter; see Example A below.

Example 1: {`outline_block`}. The outline step calls the model once per subject and produces a subject-tailored JSON section list. The output is injected verbatim into the elicitation prompt; section titles are used *exactly* as `== Heading ==` markers and the model is forbidden from adding, removing, or reordering them. This enforces structural consistency across models and experimental conditions while allowing content that adapts to the subject type-a historical figure yields biography-oriented sections; a chemical compound or a city would yield an entirely different structure.

Subject: *Vannevar Bush* **Regime:** general-domain

Filled {`outline_block`}:

```
{"sections": [
"Early Life and Education",
"Engineering Career and Raytheon",
"Wartime Science Leadership and the
OSRD",
"As We May Think and the Memex",
"National Science Foundation",
"Legacy and Influence"
]}
```

Subject: *Hammurabi* **Regime:** topic-focused ({`root_subject`} = *Ancient Babylon*)

Filled {`outline_block`}:

```
{"sections": [
"Rise to Power in Babylon",
"Military Campaigns and Territorial
Expansion",
"The Code of Hammurabi",
"Administrative and Economic Reforms",
"Role within Ancient Babylon",
"Death and Legacy"
]}
```

In topic-focused runs, the outline step is instructed to include one section explicitly connecting the subject to {`root_subject`}; here, "Role within Ancient Babylon" fulfils that requirement.

Example 2: {`phrases_block`}. After article generation, all `[[wikilinks]]` are extracted via regex and passed to the NER stage as a newline-separated list. The NER prompt receives this list as {`phrases_block`} and must classify each phrase as a true named entity deserving standalone expansion or a reject. The example below illustrates the funnel: generic role labels and loop forms are rejected; proper named entities with encyclopedic scope are accepted.

Subject: *Vannevar Bush* **Strategy:** baseline

Filled {phrases_block} (excerpt):

```
engineer
inventor
science administrator
Office of Scientific Research and Development
World War II
As We May Think
memex
Tufts University
Massachusetts Institute of Technology
History of Vannevar Bush
```

Expected NER output (baseline):

```
{"phrases": [
{"phrase": "engineer", "is_ne": false},
{"phrase": "inventor", "is_ne": false},
{"phrase": "science administrator", "is_ne":
false},
{"phrase": "Office of Scientific Research and
Development", "is_ne": true},
{"phrase": "World War II", "is_ne": true},
{"phrase": "As We May Think", "is_ne": true},
{"phrase": "memex", "is_ne": true},
{"phrase": "Tufts University", "is_ne": true},
{"phrase": "Massachusetts Institute of
Technology",
"is_ne": true},
{"phrase": "History of Vannevar Bush", "is_ne":
false}
]}
```

Generic roles ("engineer", "inventor", "science administrator") and loop forms ("History of Vannevar Bush") are mandatory rejects; all surviving entities are proper named subjects with encyclopedic scope. Under the calibrated strategy each entry additionally carries a "confidence" score and entries below $\tau = 0.75$ are rejected before reaching the similarity stage.

Example 3: {persona_block}. The persona string is injected at the system level of *every* pipeline stage-elicitation, NER, and similarity arbitration-so that editorial stance propagates uniformly into article prose, entity selection, and disambiguation decisions alike. This design isolates the effect of framing from structural generation quality and enables the controlled persona comparison reported in §5.2.

scientific_neutral (default):

```
Write in a neutral, evidence-based
register. Acknowledge uncertainty
where it exists. Prefer empirically
grounded claims. Avoid advocacy
language.
```

left_leaning:

```
Foreground issues of structural
inequality and marginalized
perspectives. Emphasize social
impact and access. Frame developments
in terms of equity, power, and
underrepresented voices.
```

conservative:

```
Emphasize institutional continuity,
traditional values, and established
social order. Frame developments in
terms of stability, heritage, and the
preservation of proven structures.
```

All three persona variants share identical structural prompts; only the framing and evaluative register differ. Because the persona is injected at every stage, it influences not only article prose but also what entities the NER stage judges encyclopedically noteworthy, enabling LLMPedia to probe how the same parametric knowledge is selected and framed under different editorial orientations.

A.1 General-Domain Prompts

Elicitation - Baseline.

System: You are LLMPedia, an elite encyclopedia writer.
Persona: {persona_block}.
Write about {subject_name} (~{avg_words_per_article} words).
STRUCTURE: Start with {{Infobox ...}} if appropriate. First text line: '''{subject_name}'''. Follow with 2–4 sentence lead. Use EACH outline title exactly as == Heading == in order.
WIKILINKS (CRITICAL): Famous subjects: 50–100 distinct [[links]]; lesser-known: 15–30. Minimum 3–5 links per sentence, woven into prose naturally. Only proper nouns: people, organizations, places, events, battles, treaties, works, awards, laws, institutions. *Forbidden:* generic concepts ("government", "economy", "education", "military"), common nouns, disciplines.
NO LOOPS / ALIASES (STRICT): Never link {subject_name} or variants. Never link [[X of {subject_name}]], [[{subject_name}'s X]], [[Part of {subject_name}]], [[History of {subject_name}]].
No <ref>, no URLs, no References section. Optional [[Category:...]].

User: Subject: {subject_name}
Section titles (use exactly):
{outline_block}

NER Filtering - Baseline.

System: You are LLMPedia's NER module. Your role: capture which subjects deserve their own encyclopedia article. Persona: {persona_block}.
ACCEPT ONLY IF BOTH: (1) True named entity (person, organization, place, event, work, law, award, institution), *and* (2) Worthy of a standalone encyclopedia entry-notable enough that readers would want to learn about it.
REJECT: {subject_name} itself, any alias, or variant. Loop patterns: "X of {subject_name}", "{subject_name}'s X", "History of {subject_name}", "Part of {subject_name}". Generic nouns, roles, concepts, dates, filler.
Output exactly: {"phrases": [{"phrase": "<exact input>", "is_ne": true/false}, ...]}

User: Subject: {subject_name}
Candidates:
{phrases_block}

NER Filtering - Calibrated.

System: You are LLMPedia's NER module. Persona: {persona_block}.
ACCEPT ONLY IF BOTH: (1) True named entity (person, organization, place, event, work, law, award, institution), *and* (2) Worthy of a standalone encyclopedia entry.
REJECT: {subject_name} itself, any alias, or variant. Loop patterns, generic nouns, roles, concepts, dates, filler.
CALIBRATED OUTPUT: Include a score (0–1) per candidate reflecting confidence that it is (a) a true named entity *and* (b) deserving a standalone article. 0.95–1.00 extremely confident; 0.85–0.95 confident; 0.75–0.85 plausible but uncertain; below 0.75 = reject.
Output exactly (valid JSON, no extra keys, no prose): {"phrases": [{"phrase": "<exact input>", "is_ne": true/false, "confidence": 0.0}, ...]}

User: Subject: {subject_name}
Candidates:
{phrases_block}

A.2 Topic-Focused Prompts

Elicitation - Calibrated.

System: You are LLMPedia, an elite encyclopedia writer. Persona: {persona_block}.
Write about {subject_name} (∼{avg_words_per_article} words).
STRUCTURE: Start with {{Infobox ...}} if appropriate. First text line: '''{subject_name}'''. Follow with 2–4 sentence lead. Use EACH outline title exactly as == Heading == in order.
CALIBRATED WIKILINKS (CRITICAL): Famous subjects: 50–100 distinct [[links (score)]]; lesser-known: 15–30. Minimum 3–5 links per sentence. Format *must* be [[Proper Noun (score)]], e.g. [[Albert Einstein (0.97)]]. Score = confidence that the entity belongs to the semantic neighborhood of the subject: 0.95–1.00 extremely confident; 0.85–0.95 confident; 0.75–0.85 plausible but uncertain. Only proper nouns: people, organizations, places, events, battles, treaties, works, awards, laws, institutions. *Forbidden:* generic concepts, common nouns, disciplines.
NO LOOPS / ALIASES (STRICT): Never link {subject_name} or variants. Never link loop forms (e.g. [[Churchill's speeches]], [[Life of Churchill]]).
No <ref>, no URLs, no References section. Optional [[Category:...]].

User: Subject: {subject_name}
Section titles (use exactly):
{outline_block}

Elicitation - Baseline.

System: You are LLMPedia, generating concise Wikipedia-style articles in Wikitext for a knowledge base built around ROOT TOPIC {root_subject}. Persona: {persona_block}.
Write a clean Wikipedia-style article about {subject_name} (∼{avg_words_per_article} words). First line: '''{subject_name}'''. Short lead (2–4 sentences) explaining what {subject_name} is and why it matters in the context of {root_subject}. Include {{Infobox ...}} if appropriate.
STRUCTURE: Use EACH section title exactly as == Title == in the given order. Do not add, remove, rename, merge, or reorder sections. Write professional, neutral paragraphs under every heading, focused on what is relevant to {subject_name} within {root_subject}.
WIKILINKS: Well-known subjects: 30–50 distinct [[wikilinks]]; obscure: 8–15. At least ∼70% of distinct links must be specific named entities or named works (people, organizations, institutions, companies, places, events, programs, projects, products, papers/books, awards, conferences, laws/policies). Broad umbrella concepts allowed only if central and ≤3 total. Wikilink the first mention of notable items strongly related to *both* {subject_name} and {root_subject}. Never link {subject_name}, {root_subject}, or trivial variants. No list/meta pages, no generic terms.
No <ref>, no URLs, no References section. Optional [[Category:...]].

User: Subject: {subject_name}
Section outline (one title per line):
{outline_block}

NER Filtering - Baseline.

System: You are the NER module of LLMPedia, expanding a topic graph *strictly* centered on ROOT TOPIC {root_subject}. Persona: {persona_block}.
ACCEPT ONLY IF BOTH: (1) The phrase is a strong, standalone encyclopedia subject, *and* (2) It has a direct, meaningful, non-trivial factual connection to ROOT TOPIC {root_subject}.
MANDATORY REJECTIONS (always is_ne = false): Phrase is exactly {subject_name} or {root_subject}, or any alias/rephrasing of either. Structural forms: "X of {subject_name}", "X of {root_subject}", "Part of {subject_name}", "{subject_name} in popular culture", "{root_subject} in popular culture". Literals, dates, URLs, verbose phrases.
Output exactly: {"phrases": [{"phrase": "<candidate>", "is_ne": true/false}, ...]}

User: Candidate phrases (one per line):
{phrases_block}

NER Filtering - Calibrated.

System: You are the NER module of LLMPedia, extending a topic graph *strictly* centered on ROOT TOPIC {root_subject}. Persona: {persona_block}.
ACCEPT ONLY IF BOTH: (1) The phrase is a strong, standalone encyclopedia subject, *and* (2) It has a direct, meaningful, non-trivial factual connection to ROOT TOPIC {root_subject}.
CONFIDENCE GUIDELINES: 0.95–1.00 extremely confident; 0.85–0.95 confident; 0.75–0.85 plausible. If relevance to ROOT TOPIC is weak *or* confidence $<0.75 \Rightarrow$ reject.
MANDATORY REJECTIONS: Phrase is {subject_name} or {root_subject}, or any alias/rephrasing. Structural variants: "X of {subject_name}", "X of {root_subject}", "Part of {subject_name}", "{subject_name} in popular culture", "{root_subject} in popular culture". Literals, dates, URLs, verbose phrases.
Output exactly (valid JSON, no extra keys, no prose): {"phrases": [{"phrase": "<candidate>", "is_ne": true/false, "confidence": <score>}, ...]}

User: Candidate phrases (one per line):
{phrases_block}

Elicitation - Calibrated.

System: You are LLMPedia, generating concise *calibrated* Wikipedia-style articles in Wikitext for a knowledge base built around ROOT TOPIC {root_subject}. Persona: {persona_block}.
Write a clean Wikipedia-style article about {subject_name} (∼{avg_words_per_article} words). First line: '''{subject_name}'''. Short lead (2–4 sentences) explaining what {subject_name} is and why it matters in the context of {root_subject}. Include {{Infobox ...}} if appropriate.
STRUCTURE: Use EACH section title exactly as == Title == in the given order. Do not add, remove, rename, merge, or reorder sections. Focus content on what is relevant to {subject_name} within {root_subject}.
CALIBRATED WIKILINKS: Format must be [[Entity (score)]], e.g. [[Albert Einstein (0.97)]]. Well-known subjects: 30–50 distinct scored links; obscure: 8–15. At least ∼70% must be specific named entities or named works. Broad umbrella concepts allowed only if central and $\leq$3 total. Score = confidence that the entity belongs to the semantic neighborhood of *both* {subject_name} and {root_subject}: 0.95–1.00 extremely confident; 0.85–0.95 confident; 0.75–0.85 plausible but uncertain. Never link {subject_name}, {root_subject}, or trivial variants. No list/meta pages, no generic terms.
No <ref>, no URLs, no References section. Optional [[Category:...]].

User: Subject: {subject_name}
Section outline (one title per line):
{outline_block}

B Pipeline Implementation

B.1 System Configuration and Design Decisions

Execution modes. LLMPedia supports two execution modes that trade latency against cost. In *online mode*, API calls are issued concurrently across a multi-threaded worker pool, with all pipeline steps - outline generation, elicitation, NER filtering, and similarity arbitration - running in parallel and a global concurrency cap enforced across them to prevent rate-limit exhaustion. In *batch mode*, requests are submitted to the OpenAI Batch API, which processes them asynchronously at a 50% cost reduction with a completion window of up to 24 hours. The two modes are equivalent by construction: they issue the same prompts under the same parameters and differ only in how requests are dispatched.

The GPT-5-mini run began in batch mode for cost efficiency. During peak expansion the Batch API returned intermittent errors, delayed completions, and batches that finished without being reported as complete, and the resulting backlog made batch dispatch impractical for the two highest-volume steps. We therefore moved NER filtering and similarity arbitration to the online API partway through that run, leaving article generation in batch mode. The open-weight runs are served on-premise and use online mode throughout. Appendix G reports the residual count differences this transition left in the logs.

Model configuration and step independence. Each pipeline step - self-grounding, outline generation, article elicitation, NER filtering, and similarity arbitration - can be assigned an independent language model, temperature, and token budget. In practice, all steps within a given run share a single model unless explicitly differentiated. The one principled exception is the embedding model used for similarity deduplication, which is always configured independently: embedding and completion models serve different functions, and conflating their configuration would silently degrade deduplication quality.

BFS depth and article budget. Expansion depth and total article count are both configurable hard caps. In **topic-focused runs** (§5.2), depth is capped at two hops from the thematic seed to maintain topical coherence. In **general-domain expansion** (§5.3), no depth cap is imposed: GPT-5-mini reaches hop 6 over ∼1M articles (1,008,947 generated), with 95.9% of filtered-in subjects at hop 4 or beyond (Table 16). In both modes, the seed article at hop 0 is generated but excluded from evaluation metrics, so that the trivially known anchor entity does not confound the measurement.

Similarity threshold and arbitration. The cosine similarity threshold above which two entity embeddings trigger a deduplication check is set to 0.90. This value is conservative: surface-form variants of the same entity (e.g. *United States* and *USA*) consistently fall above it, while genuinely distinct but thematically related entities consistently fall below it. When the threshold is exceeded, a secondary LLM arbitration step is invoked rather than rejecting automatically. This two-stage design guards against both false positives and false negatives.

NER confidence and elicitation filtering. Under the calibrated prompt variants, both the NER and elicitation steps assign explicit confidence scores. Candidates below the NER confidence threshold are rejected before reaching similarity arbitration. Rejection at either step is non-permanent: if a future parent article re-proposes the same entity with sufficient confidence, it re-enters the pipeline from the beginning.

Persona. The active persona is prepended to the system message of every LLM call in a topic-focused run - outline, elicitation, wikilink proposal, and NER filtering. Three personas are provided: scientific-neutral, left-leaning, and conservative. All three share identical structural prompts; only the framing and evaluative language differ (Appendix A).

Fault tolerance and strict-gate semantics. Failed API calls are retried with exponential backoff up to a configurable maximum. Once retries are exhausted, a strict-gate policy applies: a step that cannot produce a valid output causes the corresponding candidate to be *dropped* rather than passed through with a degraded result. For NER, a parse failure produces no candidates rather than passing all candidates unchecked; for similarity arbitration, an API failure causes the candidate to be treated as a duplicate and rejected. One consequence is visible in the funnel: a candidate dropped this way leaves no decision row, so it appears in the stage totals but not in any reject-reason breakdown (Appendix G). The policy introduces a small downward bias on recall but removes the risk of admitting candidates that bypassed quality controls.

B.2 Deduplication Correctness Under Parallel Execution

LLMPedia maintains five independent parallel queues: an *outline queue*, an *elicitation queue*, a *NER queue*, a *similarity queue*, and the *canon queue*. A subject leaves the canon queue, is outlined, then written, and its wikilinks pass to NER and similarity before any surviving child returns to the canon queue. Workers on each queue operate concurrently and independently, which creates a structural race: multiple NER workers can extract and accept the same entity from different parent articles within the same BFS wave, before any of them reaches the similarity worker. Figure 5 shows a concrete instance and its resolution. The outline and elicitation queues do not affect the argument below, since commitment happens only at Stage 3.

The main article queue is structurally duplicate-free. CANONQUEUE is the *only* channel through which an entity can reach article generation. An entity enters CANONQUEUE if and only if it simultaneously passes Stage 3 similarity *and* is absent from CANONKEYS; both operations execute atomically.

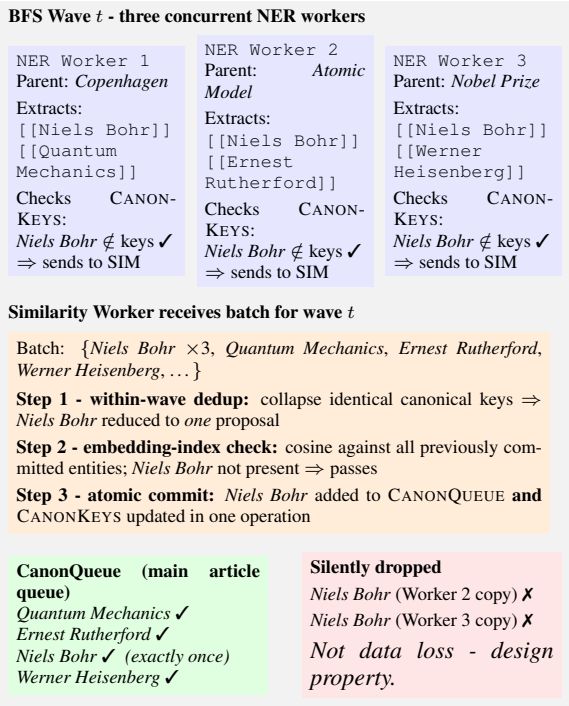


Figure 5: Three concurrent NER workers independently accept `[[Niels Bohr]]` from different parent articles. Within-wave deduplication collapses the three proposals to one before any commit.

Why commitment happens after similarity, not after NER. A candidate is registered in CANONKEYS only after it passes Stage 3. If commitment happened at NER acceptance, an entity later rejected by similarity would be permanently suppressed even though it was never materialised. Deferring commitment keeps future parent articles free to re-propose the same entity.

Within-batch collisions. A single parent article may produce the same wikilink in two surface forms that canonicalize identically. Within-wave cross-candidate deduplication handles this before any index update.

Correctness guarantee. By induction on BFS waves, the embedding index at the start of wave t contains exactly the entities committed in waves $0, \ldots, t-1$. The main article queue therefore contains no duplicate entities at any point during or after execution. The cost of enforcing this under parallelism is the post-similarity loss quantified in Appendix G.7.

C Evaluation Prompt

C.1 Claim Extraction Prompt

System: You are a factuality evaluator. Extract up to $\{K_{\max}\}$ distinct, atomic, verifiable factual claims from an article about "{subject}". Each claim must be ONE atomic fact (one predicate, one relationship), self-contained (include the subject name), and verifiable against a reference encyclopedia. Skip opinions, hedged language, and vague statements. Output ONLY valid JSON: {"claims": ["claim1", "claim2", ...]}

User: Subject: {subject} Article text: {article_text} Return JSON with up to $\{K_{\max}\}$ claims.

C.2 Factuality Verification Prompt

This is the verification prompt used for all results reported in this paper. The absence-of-evidence asymmetry is encoded as a hard rule with three worked examples; without this rule a naive judge conflates "evidence does not mention X" with "evidence refutes X", mechanically inflating the false-claim rate and obscuring the distinction between knowledge gaps and factual errors that is central to our analysis.

System: You are a careful fact-checker. Verify {n} claims about "{subject}" against the evidence below.
Use exactly one of these three verdicts per claim:
"supported": the evidence explicitly states the claim, or the claim follows directly from what the evidence says.
"refuted": the evidence explicitly states something that CONTRADICTS the claim. There must be a positive statement in the evidence that is logically incompatible with the claim (e.g. claim says "founded in 1850" but evidence says "founded in 1920").
"insufficient": the evidence does not address the claim, addresses it only partially, or is silent on the topic. THIS INCLUDES every case where the evidence simply does not mention the subject of the claim.
CRITICAL RULE: Absence of evidence is NOT evidence of absence. If the evidence does not mention X, the verdict for any claim about X is "insufficient", NEVER "refuted". Mark "refuted" ONLY when the evidence contains an explicit positive statement that contradicts the claim.
EXAMPLES:
Claim: "The school was founded in 1920."
Evidence: "The school was founded in 1850."
Verdict: *refuted* (evidence states a different founding year)
Claim: "The school was founded in 1920."
Evidence: "The school is located in Paris and offers art programs."
Verdict: *insufficient* (evidence does not address the founding year)
Claim: "The curriculum was influenced by the Bauhaus."
Evidence: "There is no mention of Bauhaus or curriculum influences."
Verdict: *insufficient* (absence of mention is not contradiction)
When verdict is **"refuted"**, additionally output an `error_type` field with one of these values: `temporal` (wrong date/year/era), `numerical` (wrong quantity), `spatial_geographic` (wrong location), `person_attribution` (wrong person credited), `entity_relation` (wrong relationship), `causal_motivational` (wrong cause/consequence), `categorical_taxonomic` (wrong classification), `definitional_property` (wrong intrinsic property), `fabricated_entity` (entity does not exist), `partial_truth_overgeneralization` (partly correct but overstated), `terminological_naming` (wrong name/label), `other`. Set `error_type` to null for supported or insufficient.
Output ONLY valid JSON:

```
{"verdicts":[
{"idx":1,"verdict":"...",
"error_type":"...",
"confidence":0.0--1.0,
"explanation":"..."}
]}
```

User: Claims:
{claims_block}
Evidence source: {evidence_source}
{evidence_snippets}
Return JSON with verdicts.

Parser safety net. The parser additionally coerces any verdict labeled *refuted* whose explanation contains silence-indicating phrases (*does not mention*, *not specified*, *there is no indication*) into *insufficient*, logging each coercion. This affected 2.3% of initially refuted verdicts across our runs.

D Web Evidence Pipeline

The pipeline auto-detects available search backends (Valyu → Serper → Brave → DuckDuckGo); Valyu was the primary backend in all large-scale experiments.

Admission is by rule, not by list. Each candidate URL receives a quality score on a 0–100 scale and is eligible for full-page fetching only at ≥ 60. Any institutional or editorially accountable source meets the rule: unknown `.edu`, `.gov`, and `.org` domains are admitted at a default score of 70 whether or not we have encountered them before, which is how the tier reaches agencies, university departments, museums, and archives we never enumerated. Unknown HTTPS domains default to 35 and are excluded, as are the root domains in Table 9.

Tables 10–11 list the **133 domains we scored by hand**. These calibrate the scale and illustrate the categories the rule is meant to capture - encyclopedias, government, wire services, academic journals, museums and libraries, history, science, universities, fact-checking and data, legal, and medical. They are not the set of sources the pipeline will accept: most fetched sources are admitted by the default rule above, and a source absent from these tables is not thereby excluded.

D.1 Frontier Results Summary

Of 433 Wikipedia-absent subjects in the 1,000-article random sample, 311 (71.8%) yielded usable web evidence. Precision 98.3%, true rate 57.6%, false rate 0.6%, unverified 41.8%. The remaining 28.2% are excluded from Tier 2 frontier statistics - a source-selection bias discussed in §7.

E Human Validation Protocol

E.1 Design

We ran a two-stage human validation to audit the LLM judge against expert human verdicts.

Sampling. For each evaluation tier, we drew a balanced stratified sample from the judge's verdict distribution. **Tier 1 (Wikipedia evidence)**: 99 (claim, evidence, judge verdict) triples, exactly balanced as **33 supported / 33 refuted / 33 insufficient**, drawn from the random subsample of the 1M GPT-5-mini corpus. **Tier 2 (curated web evidence)**: 30 triples, balanced as **10 / 10 / 10**, drawn from the frontier-Web subset.

Category	Blocked root domains
Circular / Wikimedia	wikipedia.org, en.wikipedia.org, en.m.wikipedia.org, wikidata.org, www.wikidata.org, wikimedia.org
Search engines	google.com, bing.com, duckduckgo.com, yahoo.com
Social / video	facebook.com, instagram.com, twitter.com, x.com, tiktok.com, reddit.com, quora.com, youtube.com, pinterest.com
E-commerce	amazon.com, ebay.com
AI answer engines	perplexity.ai

Table 9: Root-domain exclusions. Exclusion uses root-domain matching, so subdomains of blocked roots are also excluded. Wikimedia domains are blocked so that Tier 2 cannot recirculate Tier 1 evidence.

Category	Domain	Score	Domain	Score	Domain	Score
Encyclopedias	britannica.com	100	worldhistory.org	90	encyclopedia.com	85
	scholarpedia.org	88	plato.stanford.edu	95	iep.utm.edu	88
	newworldencyclopedia.org	75				
Government	loc.gov	97	archives.gov	97	congress.gov	95
	usa.gov	93	cia.gov	90	state.gov	90
	whitehouse.gov	90	nasa.gov	95	nih.gov	94
	cdc.gov	93	fda.gov	90	epa.gov	89
	noaa.gov	90	usgs.gov	90	nps.gov	85
	si.edu	92	parliament.uk	88	gov.uk	87
	europarl.europa.eu	85	un.org	88	who.int	90
	worldbank.org	88	imf.org	87		
Wire services	reuters.com	94	apnews.com	94	bbc.com	93
	bbc.co.uk	93	nytimes.com	92	washingtonpost.com	90
	theguardian.com	91	economist.com	90	ft.com	89
	wsj.com	89	theatlantic.com	87	newyorker.com	87
	npr.org	88	pbs.org	88	aljazeera.com	85
	dw.com	84	france24.com	83		
Academic journals	nature.com	95	science.org	95	sciencedirect.com	90
	springer.com	89	link.springer.com	89	wiley.com	88
	onlinelibrary.wiley.com	88	tandfonline.com	87	cell.com	93
	thelancet.com	93	bmj.com	92	nejm.org	94
	pnas.org	92	ncbi.nlm.nih.gov	92	pubmed.ncbi.nlm.nih.gov	92
	jstor.org	90	arxiv.org	85	ssrn.com	82
	researchgate.net	75	scholar.google.com	80*	doaj.org	80
	semanticscholar.org	82	ieee.org	89	ieeexplore.ieee.org	89
	acm.org	88	dl.acm.org	88		

Table 10: Hand-scored domains (Part I). These calibrate the 0–100 scale and illustrate the categories that qualify; a domain absent from this table is not thereby excluded, since unknown .edu/.gov/.org domains are admitted at 70. *scholar.google.com is scored but excluded at runtime because google.com is blocked by root-domain matching.

Annotation. For each (claim, evidence, verdict) triple, a human expert was shown the full evidence snippets used by the judge and annotated: (a) do you agree with the judge verdict? and (b) if not, which of the three classes do you believe is correct?

E.2 Results

Aggregate conservation. Per-claim agreement is imperfect, but disagreements largely cancel in aggregate. From a balanced judge sample of 33/33/33 (Tier 1), human class totals are 37/24/38; from a balanced 10/10/10 (Tier 2), totals are 13/7/10. The *supported* total-which determines the headline true rate-drifts by only $+4$ of 33 (Tier 1) and $+3$ of 10 (Tier 2), so the macro estimate is robust to per-claim variation. This conservation is what allows the 68.4% headline true rate and the 30.5% unverifiable rate to be reported with confidence despite imperfect per-claim agreement.

Discussion. Refuted carries the lowest agreement, which is expected: the refuted class is the one where human and judge most often disagree about whether the evidence positively contradicts the claim or merely fails to mention it. The verification prompt (Appendix C) addresses this by stating the absence-of-evidence asymmetry as a hard rule and including three worked examples; the parser additionally coerces any verdict labelled refuted whose explanation contains silence-indicating phrases (*does not mention*, *not specified*, *there is no indication*) into insufficient, which affected 2.3% of initially refuted verdicts.

Category	Domain	Score	Domain	Score	Domain	Score
Museums / Libraries	smithsonianmag.com	87	nationalgeographic.com	86	metmuseum.org	90
	moma.org	87	nga.gov	88	bl.uk	90
	bnf.fr	88	dpla.org	83	europeana.eu	83
History / Biography	history.com	80	biography.com	75	oxforddnb.com	90
	anb.org	88	historytoday.com	78	historyextra.com	77
Science / Technology	scientificamerican.com	86	newscientist.com	83	livescience.com	75
	space.com	76	phys.org	78	sciencenews.org	82
	quantamagazine.org	88	arstechnica.com	80	spectrum.ieee.org	84
	technologyreview.com	84				
Universities	mit.edu	93	stanford.edu	93	harvard.edu	93
	ox.ac.uk	92	cam.ac.uk	92	berkeley.edu	91
	caltech.edu	91	yale.edu	91	princeton.edu	91
	columbia.edu	90	uchicago.edu	90	cornell.edu	89
	cmu.edu	89	ethz.ch	89	mpg.de	89
	khanacademy.org	80				
Fact-checking / Data	snopes.com	82	factcheck.org	84	politifact.com	80
	ourworldindata.org	88	statista.com	78	data.gov	88
	census.gov	90	bls.gov	89	bea.gov	88
Legal	law.cornell.edu	90	supremecourt.gov	92	courtlistener.com	80
	oyez.org	82				
Medical / Health	mayoclinic.org	88	clevelandclinic.org	85	webmd.com	72
	medlineplus.gov	90	hopkinsmedicine.org	87	uptodate.com	88

Table 11: Hand-scored domains (Part II). With Table 10 these are the **133** domains carrying hand-assigned scores. They are examples of what the admission rule accepts, not its extent: any `.edu`, `.gov`, or `.org` domain outside these tables is admitted at the default score of 70.

Class	Tier 1 (Wiki)	Tier 2 (Web)
Supported	87.9%	93.3%
Refuted	81.8%	70.0%
Insufficient	93.9%	90.0%
Overall	87.9%	84.4%

Table 12: Human–judge per-class agreement. Agreement on the contribution-bearing classes (*supported* drives true rate; *insufficient* drives unverifiable rate) ranges **87.9–93.9%** across both tiers, with a per-class floor of 87.9%.

F Ablation: Full Results

We conduct a 2×2×2 factorial ablation over **self-grounding** (on/off), **prompting strategy** (baseline vs. calibrated), and **reasoning budget** (minimal vs. low) on `gpt-5-mini`. For each of the eight configurations, we generate 5K articles and evaluate a fixed random sample of 500 articles (seed 42). Claims are verified against full Wikipedia pages by `gpt-4.1-nano`.

Prompting strategy. Switching from baseline to calibrated yields only small and inconsistent gains. Without self-grounding, precision changes from 94.7% to 94.5%; with self-grounding, from 96.2% to 96.3%. The main effect is a modest reshaping of the true–unverifiable balance, not a large precision jump.

Self-grounding. Self-grounding improves results consistently. Under calibrated+minimal, precision rises from 94.5% to 96.3% and the unverifiable rate falls from 5.8% to 2.9%. Its main benefit is reducing unverifiable claims. The cost is substantial: approximately 33% higher generation cost (∼$850 more per million articles).

Reasoning budget. Increasing the reasoning budget from minimal to low *reduces* precision in all four SG×prompt settings. Longer reasoning traces introduce extra speculative detail rather than improving factual reliability.

Configuration choice. The best configuration is **SG + calibrated + minimal** at 96.3% precision and 93.7% true rate. We choose **no SG + baseline + minimal**† for all large-scale runs: it saves ∼$850 per million articles while sacrificing only 1.6 pp precision.

G Entity Sanitization Funnel: Full Breakdown

This appendix presents the full sanitization funnel for all three models: cross-model comparison figures (Appendix G.1), per-model deep dives (Appendices G.2–G.4), pipeline coverage statistics distinguishing the filtered-in corpus from the expanded-out subset (Appendix G.5), the alias absorption introduced by Stage 3 disambiguation (Appendix G.6), and the residual gap between

SG	Prompt	Reas.	Prec.%	True%	False%	Unv.%
✗	base	low	90.7	84.6	8.2	7.3
✗	base	min	94.7†	89.5	4.5	6.0
✗	calib	low	92.1	85.4	7.0	7.7
✗	calib	min	94.5	89.7	4.5	5.8
✓	base	low	93.9	89.3	5.5	5.2
✓	base	min	96.2	93.0	3.6	3.4
✓	calib	low	95.3	92.2	4.3	3.6
✓	calib	min	**96.3**	**93.7**	**3.4**	**2.9**

Table 13: Full 2^3 ablation on `gpt-5-mini` (500 articles, seed 42). SG = self-grounding. †: selected operating point.

similarity acceptance and queue insertion (Appendix G.7).

Counting conventions. Four quantities in this appendix are not the same and are not meant to be. *Sim-accepted* counts candidates that passed Stage 3 arbitration. *Inserted* counts those that then won the atomic commit, so the difference between the two is the loss reported in Appendix G.7. *Filtered-in* counts members of the corpus, which adds the seed article and drops rows left unprocessed when a run hit its ceiling. *Total* is every entity row the run touched. Each table states which of the four it reports. Survival rates are stage-relative: canonical survival is measured against pre-NER input, NER survival against canonical-kept, and similarity survival against NER-accepted; the cross-model figures in Appendix G.1 instead plot rates relative to raw candidates, so their canonical values are lower. Hop indices also differ by one between the two views: those figures label candidates by the wave that produced them, whereas the coverage tables label each subject by its own hop, which is one greater.

Operational note. Small residuals remain after these conventions are applied, and they are largest for GPT-5-mini because that run is an order of magnitude bigger and ran far longer: ∼1M articles against 120K. Two things happened over that period. Jobs on the shared cluster were preempted and resubmitted mid-wave, and the OpenAI Batch API returned intermittent errors and delayed completions during peak expansion, including batches that had finished but were never reported as such. Queue overflow eventually made batch mode impractical for the NER and similarity stages of the GPT-5-mini run, so we moved those calls to the online API partway through. Resubmitting work after these interruptions re-ran entities that had already been processed; the duplicates went to canonical deduplication and Stage 3 arbitration and were removed there rather than entering the corpus. Two visible traces remain in this appendix: 85,684 GPT-5-mini candidates (3.74%) reached NER without producing a decision row, against 538 for DeepSeek and 457 for Llama; and 5.08% of GPT-5-mini's similarity-accepted candidates did not reach the queue, against 0.04% and 0.20%. Both are removals rather than additions, and neither reaches the factuality samples, which are drawn from the committed corpus after deduplication.

	GPT-5-mini	DeepSeek	Llama
Generated articles	1,009K	120K	120K
Raw `[[links]]`	70.2M	6.5M	9.8M
Pre-NER candidates	12.48M	2.00M	4.33M
After canonical dedup	2.29M	859K	1.25M
After NER	1.65M	444K	684K
After similarity	1.12M	396K	499K
New queued subjects	1.06M	396K	498K
Raw → pre-NER	17.8%	30.8%	44.1%
Canonical dedup survival	18.4%	42.9%	28.8%
NER survival	72.1%	51.7%	54.8%
Sim. survival	67.8%	89.3%	73.0%
Raw → queue	1.52%	6.10%	5.07%

Table 14: Full funnel with per-stage survival rates across all three models. Survival is stage-relative (see counting conventions). Raw → pre-NER reflects article-level candidate construction and surface-form consolidation before any LLM stage runs. Canonical deduplication is dominated by previously committed entities: as the committed index grows, canonical survival falls and canonical deduplication loss increases.

G.1 Cross-Model Comparison

Figures 6–10 present the five cross-model figures generated by the combined funnel analysis.

Reading the curves together. Figures 8–10 tell a consistent story across all three models: the funnel becomes strictly more selective at each successive hop, not because the filters become harsher (their per-call behavior is nearly stationary) but because the population entering the filters becomes proportionally more redundant. This is the structural reason the hop-stratified factuality results in Table 6 attach to a thinning external-evidence tail rather than to deteriorating model knowledge.

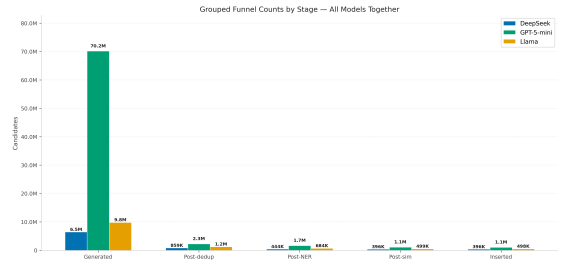


Figure 6: Funnel counts per stage, grouped by model. Each cluster on the x-axis is a pipeline stage; bars within a cluster are the three models. The decay from *Generated* to *Inserted* is visible as a near-monotone right-to-left shrinking pattern for every model. GPT-5-mini operates at roughly an order of magnitude larger scale at every stage, but the *shape* of the funnel is remarkably similar across models.

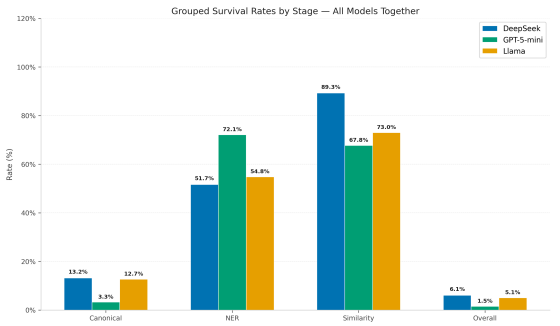


Figure 7: Per-stage survival rates, grouped by model. Rates here are relative to raw candidates and so differ from the stage-relative rates in Table 14; the ordering is unchanged under either denominator. Three patterns stand out: (i) GPT-5-mini has the lowest canonical survival (3.3% raw-relative, 18.4% stage-relative) because its much larger committed index makes more candidates redundant; (ii) Llama's NER survival is the lowest at 54.8% and is dragged down by parse failures rather than legitimate "not a named entity" rejections (Appendix G.4); (iii) DeepSeek's similarity survival is the highest at 89.3%, indicating its candidate set carries less semantic redundancy at the post-NER stage.

G.2 GPT-5-mini Deep Dive

Total: 1,063,949 entities; 1,008,947 with articles. Only 27 entities (0.003%) had articles generated but were never processed through the full pipeline.

G.3 DeepSeek-V3 Deep Dive

Total: 396,091 entities; 120,139 with articles. The run produced articles at hops 0–5 with 37 entities (0.03%) showing partial processing.

G.4 Llama-3.3-70B Deep Dive

Total: 498,226 entities; 120,100 with articles. 3 entities (0.003%) showed partial processing. The Llama run also has the largest NER-parse-failure footprint, discussed below.

NER parse failures. The 270,157 Llama parse failures in Figure 24 are the single largest deviation between models. Inspection of the per-chunk parse mode log (`ner_decisions.jsonl`) shows that 99.7% of NER *calls* were parsed as native JSON; the remaining 0.3% recovered via regex-based salvage or were treated as exceptions. The strict-gate policy in `factuality_core` rejects every candidate inside a chunk whose model output did not produce a usable decision row, which inflates the reject count without representing a quality judgement about the candidate. This is a conservative but blunt mechanism: a lower-compliance model's effective NER recall is slightly underestimated, but the corpus is protected from candidates that bypassed the filter. Llama's downstream similarity stage and inserted-corpus quality remain comparable to the other two models (Figure 25, Table 14).

Model	Total	Filt.-in	Exp.-out	Terminal
GPT-5-mini	1,063,949	1,063,930	174,100	83.6%
DeepSeek	396,091	396,091	35,441	91.1%
Llama-70B	498,226	498,226	52,557	89.5%

Table 15: Pipeline coverage at corpus level. Terminal percentage is the share of total entities that were filtered in but not expanded out. The high terminal share reflects each run reaching its computational ceiling at the deepest hop, not pipeline failure.

G.5 Pipeline Coverage: Filtered-In vs. Expanded-Out

A subject in the LLMpedia corpus has two distinct lifecycle states. A subject is *filtered-in* once it has passed Stage 1–3 sanitization as a *candidate* produced by some parent article - this is the precondition for receiving an article of its own. A subject is *expanded-out* once NER has run on its own outbound wikilinks, producing candidates for the next BFS hop. Every expanded-out subject is filtered-in by construction; the converse does not hold. The gap between the two corresponds to subjects whose article was generated but whose outbound expansion did not run, either because the BFS reached its hop or subject-count limit, or because the run was stopped.

Why this distinction matters. The headline factuality numbers in Table 6 and the cross-model

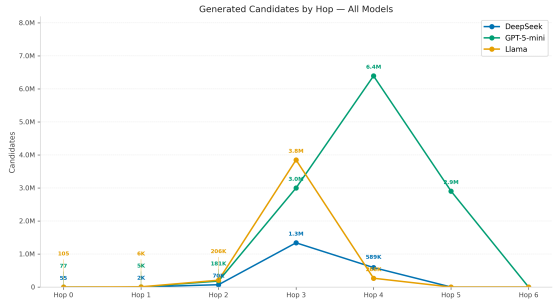


Figure 8: Pre-NER candidate count by BFS hop, one line per model. The characteristic peak-and-decay shape reflects two regimes: rapid growth as BFS reaches the wide middle hops (peaking at hop 3 for the open-weight models and hop 4 for GPT-5-mini), followed by collapse once most surfaced entities are already in the committed index and most parents are pruned at canonical dedup.

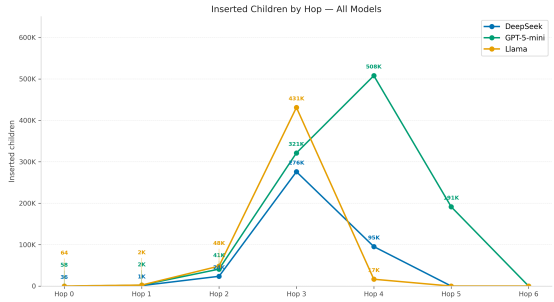


Figure 9: Inserted children per hop, one line per model, indexed by the wave that produced them. This is the shape of the resulting corpus: the bulk of GPT-5-mini's ∼1M articles come from waves 3–4 (321K and 508K), which appear as hops 4–5 in Table 16. The open-weight runs terminate two waves earlier because the 120K max-subject budget was reached. The shared asymmetry - light early waves, dense middle waves, depleted beyond - is characteristic of BFS over a power-law-degree encyclopedic graph.

comparison in Table 5 sample from *filtered-in* subjects: these are full articles, vetted by NER and similarity, and constitute the actual LLMpedia corpus a reader would consume. The hop-survival rates in Appendix G.1 are computed over *expanded-out* subjects, since these are the only ones that produced a measurable outbound funnel. Reporting both quantities explicitly avoids two common misreadings: (i) that the corpus is smaller than reported because some entities did not fully traverse the pipeline (it is not - filtered-in articles are real members of the corpus), and (ii) that survival rates somehow apply to subjects that never received an outbound NER call (they do not).

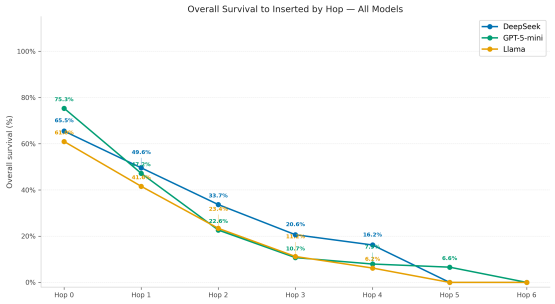


Figure 10: Overall survival from raw candidate to inserted queue subject, by BFS hop and model. Survival falls sharply with depth: for GPT-5-mini the rate drops from 75.3% at hop 0 to 6.6% at hop 5, and the open-weight models trace similar trajectories. The cause is not the LLM stages (NER and similarity survival are relatively flat with depth) but canonical dedup, whose survival collapses as the committed index saturates the local neighborhood of the BFS frontier.

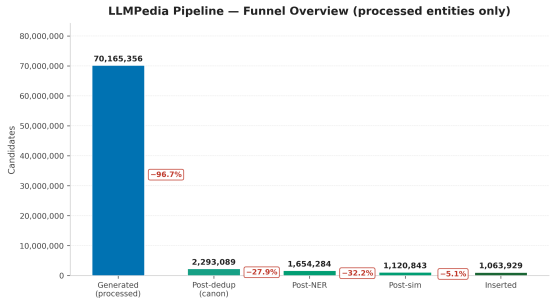


Figure 11: GPT-5-mini five-stage funnel from 70.2M raw candidates to 1.06M queued subjects. The largest absolute loss is at pre-NER canonical deduplication (−81.6% of pre-NER input; −96.7% of raw candidates, as plotted), reflecting that most wikilinks in any given article are surface variants of entities already committed elsewhere in the corpus.

G.6 Alias Absorption

Per-model totals are reported in the main text (Table 3, §5.1). Every Stage 3 similarity rejection records the canonical entity the candidate was found to duplicate, and these *alias absorptions* are the duplicate articles LLMpedia would have produced had Stage 3 been disabled. Absorption volume scales with corpus size, since a larger committed index gives each new candidate more opportunities to match something already present. Note that alias counts are lower than total Stage 3 rejections, because within-wave collapses and operational rejections are resolved before any canonical entity is recorded.

Patterns in the top-absorbing entities. Across all three models the top-30 absorbed-alias lists are

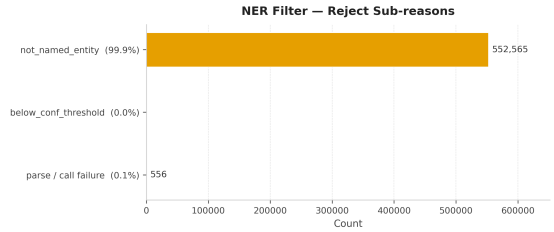


Figure 12: GPT-5-mini NER reject reason breakdown. Of 553,121 rejections carrying a recorded decision, 99.9% (552,565) are the legitimate `not_named_entity` class and 556 are parse failures. A further 85,684 candidates (3.74% of those entering NER) produced no decision row at all, which is why subtracting NER-accepted from canonical-kept in Table 14 yields a larger figure than the rejections shown here; the equivalent counts are 538 for DeepSeek and 457 for Llama, and the cause is the batch-mode interruptions described in the operational note above.

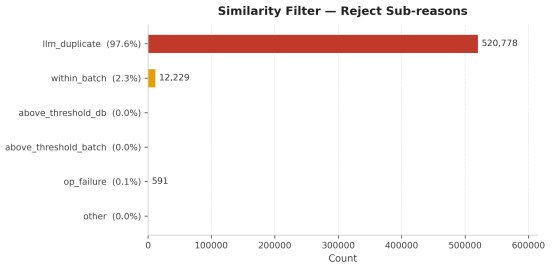


Figure 13: GPT-5-mini similarity reject reasons. Of 533,598 rejections, 97.6% (520,778) are LLM-confirmed duplicates - the arbitration LLM agreed with the embedding threshold that the candidate was a surface variant of an already-committed entity. Within-batch duplicates (2.3%, 12,229) are concurrent NER workers independently nominating the same entity in the same BFS wave (Appendix B.2). Operational failures (591, 0.1%) come from transient batch-output gaps and SDK exceptions; the strict-gate policy treats these as duplicates to avoid contaminating the corpus, at the cost of a small downward recall bias.

dominated by: (i) U.S. federal departments (variants: *U.S. Department of X* / *United States Department of X* / *Department of X (United States)* / *US Department of X*); (ii) major universities (acronym variants and *University of X* / *X University* alternations); (iii) well-known geographical locations (with and without state qualifiers); (iv) prominent international organizations (full name vs. acronym). These categories are exactly the ones where Stage 1 canonical keying cannot resolve the variant space (the strings are too different) and Stage 3 LLM arbitration is the only mechanism that can. Figures 29, 30 and 31 give the per-model lists.

Magnetic-entity caveat. A small number of entities absorb a disproportionate share of aliases. In the GPT-5-mini run, the top entity (*University of Oxford*) absorbed 1,814 aliases - more than the entire DeepSeek run produced for any single canonical entity (top: *University of Illinois at Urbana–Champaign*, 198). Manual inspection of a sample of these absorptions shows the bulk are correct collapses, but a small fraction (<2% in the sample) are arguable: e.g. DeepSeek's *Charlestown, Massachusetts* entity absorbed both *Charlestown, Boston* and the bare *Charlestown*, which is reasonable in the U.S. historical context of the article that nominated it but might in principle have justified separate disambiguation pages. Errors of this form leave the corpus slightly under-representing minor place names; they do not affect any of the factuality numbers reported in the paper.

G.7 Post-Similarity Loss

Even after Stage 3 similarity accepts a candidate, the candidate may not enter the queue. Tracing the `sim_worker_loop` in the implementation, three things can drop a similarity-accepted candidate between post-sim and inserted: a depth-cap rejection (`hop+1 > max_depth`), a max-subjects-cap rejection (the main queue refuses new rows), or a canonical-key race (two workers sim-accepted the same surface form before either committed; only one wins). For the unbounded-depth runs reported in this paper, the depth-cap cause does not apply.

Implication for the correctness claim. The deduplication-correctness argument in Appendix B.2 establishes that the main article queue is structurally duplicate-free at all times. The loss reported in Table 17 is the cost of enforcing that guarantee under parallelism: when two workers reach Stage 3 with the same canonical key in the same wave, exactly one wins the atomic commit and the other is dropped. This is bounded, one-directional, and what a correct dedup implementation must do; it is not a hidden source of corpus loss.

H Persona Analysis

This appendix supplements the persona analysis summarized in §5.2. Section H.1 describes the framing-evaluation pipeline and statistical procedure. Section H.2 reports vocabulary hit rates per

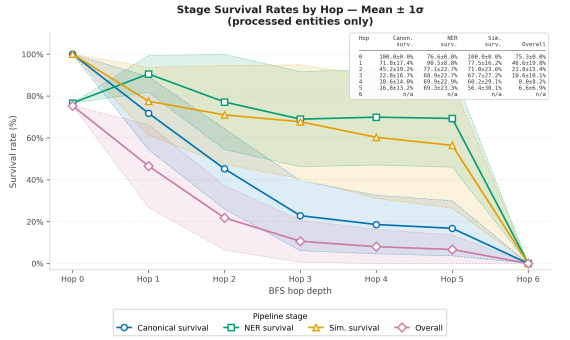


Figure 14: Per-hop survival rates for GPT-5-mini with $\pm 1\sigma$ bands across entities. Three things to read off: (i) overall survival is monotonically decreasing in hop, from 75.3% to 6.6%; (ii) variance across entities within a hop is large, especially at deep hops where some entities produce dozens of survivors and many produce zero - the power-law tail of wikilink fan-out; (iii) the NER and similarity stages stay comparatively flat (NER 90.5%→69.3%, similarity 77.5%→56.4% over hops 1–5) while canonical survival collapses from 71.8% to 16.8%, which is what drives the overall decline.

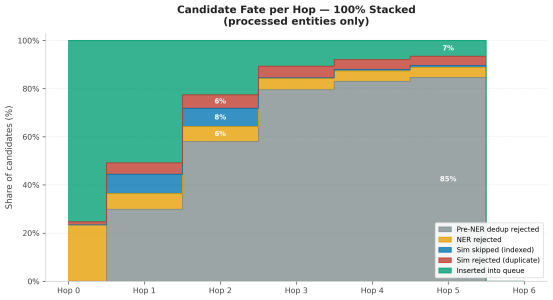


Figure 15: GPT-5-mini candidate fate per hop, 100% normalized stacked area. At hop 1 most candidates make it through the entire pipeline to insertion; by hop 5 the pre-NER dedup share dominates and the inserted share has shrunk to a few percent. The displacement is driven by the redundancy layer, not by NER or similarity rejecting more aggressively at depth.

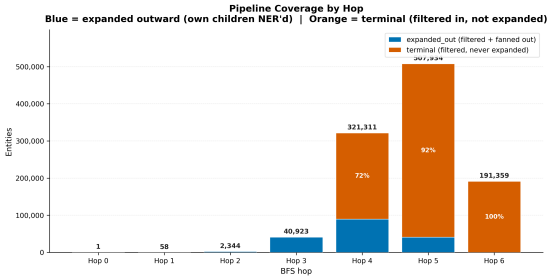


Figure 16: GPT-5-mini pipeline coverage by hop. Blue bars are *expanded-out* entities (NER ran on their outbound wikilinks); orange bars are *terminal* entities (article exists and is validly filtered into the corpus, but never expanded). Terminality is concentrated at hops 5–6 where the run reached its computational ceiling: 191,359 of the 1.06M subjects are terminal at hop 6, which is the natural BFS boundary. The factuality evaluation samples from filtered-in entities at all hops regardless of expanded status, which is why the headline coverage and true-rate numbers are reported over the full ∼1M corpus.

framing category across personas and topics, and lists every word in the framing lexicons we used. Section H.3 lists all Bonferroni-significant framing shifts on the three primary topics. Section H.4 reports the actor-selection analysis that quantifies which named individuals each persona foregrounds. Section H.5 reports structural similarity across personas (outline overlap, n-gram overlap, function-word cosine). Section H.6 reports the extended analysis on *One Piece* and *Quantum Physics*.

H.1 Methodology

Overview. The framing pipeline is entirely separate from the factuality pipeline and uses **no LLM judge**: every metric reported in this appendix is computed by a deterministic word-list classifier and standard stylometric measures. Given the same article text and the same word lists, every score is bit-identical across runs. This design choice is deliberate: the framing analysis is intended to be fully auditable, so a reviewer can read the vocabulary lists in Table 18 and decide whether they operationalize the named dimensions fairly.

Article pairs and sampling. For each (model, topic), we identify the subjects present under all three personas and random-sample 30 from their intersection (seed 42). Each sampled subject has exactly three articles - one per persona - written by the same model on the same topic. We form the three pairwise comparisons (conservative vs. left-leaning, conservative vs. scientific-neutral, left-leaning vs. scientific-neutral), giving 90 matched pairs per (model, topic). Pairing over subjects is essential: each statistical test compares the same subject written under two different personas, removing subject-level confounds.

Text preparation. Before scoring, each article's Wikitext is stripped: `[[link|display]]` yields *display*, plain `[[link]]` yields *link*, template blocks ({{...}}) and HTML tags are removed, and whitespace is collapsed. The cleaned text is tokenized by extracting all Unicode word

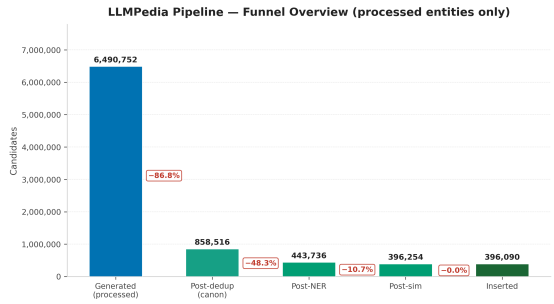


Figure 17: DeepSeek five-stage funnel from 6.5M raw candidates to 396K queued subjects. The pre-NER canonical dedup stage is less severe than for GPT-5-mini (−57.1% vs. −81.6% of pre-NER input) because the committed index is smaller; the LLM stages take a proportionally larger share of the total loss.

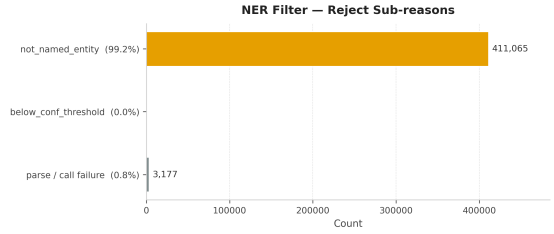


Figure 18: DeepSeek NER reject reasons. Of 414,242 rejections, 99.2% (411,065) are `not_named_entity` - the filter behaving as designed. Parse failures account for 0.8% (3,177 rejections), with the remainder recovered via the JSON-substring fallback.

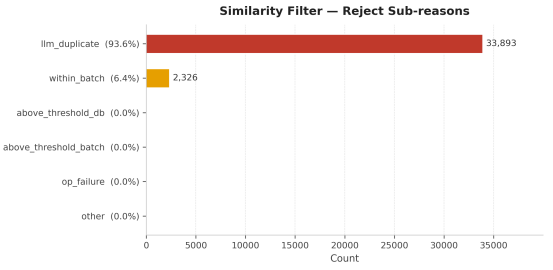


Figure 19: DeepSeek similarity reject reasons. Of 36,219 rejections, 93.6% (33,893) are LLM-confirmed duplicates and 6.4% (2,326) within-batch duplicates; no operational failures occurred during this run.

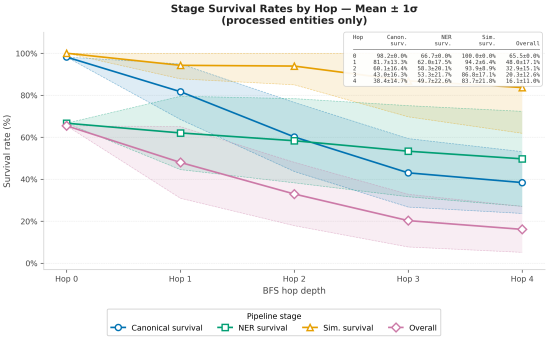


Figure 20: DeepSeek per-hop survival rates with $\pm 1\sigma$ bands. The overall survival trajectory mirrors GPT-5-mini's (65.5% at hop 0 down to 16.2% at hop 4) but with consistently higher similarity survival, reflecting the smaller embedding index.

characters in lowercase. This produces the token sequence on which all subsequent metrics operate.

Word-list framing classifier. Each article is independently scored on 24 framing dimensions by counting how often vocabulary from a fixed word list appears in the cleaned token sequence. For each framing dimension, single-word entries are matched with a token frequency counter; multi-word phrases are matched by substring search over the joined token sequence. Hit counts are normalized per 1,000 content tokens so that article-length differences do not confound the scores. The complete vocabulary for each of the 24 dimensions is given in Table 18 (domain-specific lists used on the three primary topics) and Table 23 (topic-agnostic lists used in the extended analysis). We deliberately did not expand the lists with synonyms from word embeddings or thesauri, because that would substitute opaque similarity-based recall for the interpretable surface match. Some vocabulary appears in multiple framing dimensions (e.g., *marginalized* contributes to `civil_rights_progressive`, `left_progressive`, and `decolonial_framing`; *tradition* to `epistemic_hedging`, `right_conservative`, and `indigenous_agency`). These overlaps are intentional: the framings are conceptually related and a single token can legitimately contribute to multiple scores.

Stylometric metrics. In addition to framing hits, we compute six stylometric measures per pair: outline Jaccard (set similarity of section headings), n-gram Jaccard for $n \in \{1, 2, 3\}$ (phrase overlap), function-word cosine (stylometric fingerprint on a fixed list of 100 function words), mean sentence length, type-token ratio, and sentiment net (counts of generic positive minus negative terms per 1,000 tokens). All are deterministic, language-model-free, and computable from token counts.

Actor-selection analysis. For each persona-pair, we also compute hit rates for individual named persons from the actor lexicons (e.g.

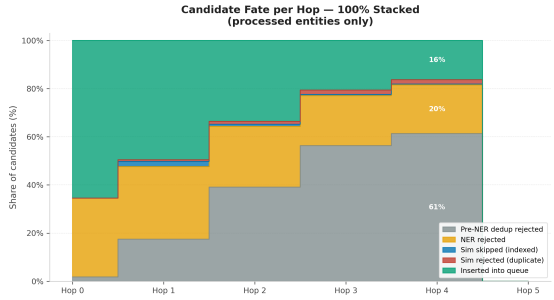


Figure 21: DeepSeek candidate fate per hop, 100% normalized. The insertion fraction is meaningfully higher at every hop than for GPT-5-mini, again because of the smaller committed index. The shape of the displacement - pre-NER dedup growing with depth - is identical to the other two models.

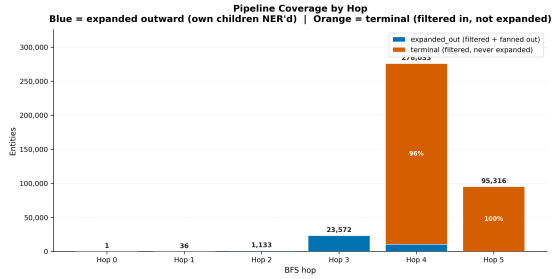


Figure 22: DeepSeek pipeline coverage by hop. The run terminated with hop 4 only partially expanded: 10,719 of 276,033 hop 4 entities (3.9%) are expanded-out, and hop 5 is entirely terminal. The 96.1% terminal share at hop 4 reflects the max-subject budget being reached: those 265K entities legitimately entered the corpus as filtered-in articles, but the pipeline was stopped before their outbound wikilinks could be processed.

`civil_rights_actors_progressive`, `civil_rights_actors_canonical`). This gives a per-actor view of which named individuals each persona foregrounds, beyond the aggregate framing-category scores.

Statistical testing. For each (model, topic, persona-pair, framing-dimension) tuple we run a paired Wilcoxon signed-rank test on the 30 matched per-article hit-rate differences, where each observation is the difference in hit rate between persona A and persona B for the same subject. We apply Bonferroni correction over all tests within each analysis (648 tests in the primary analysis; $\alpha = 0.05/648 \approx 7.7 \times 10^{-5}$) and report rank-biserial correlation r_{rb} as effect size. **37 of 648 tests** reach Bonferroni significance on the primary topics with clear directional patterns.

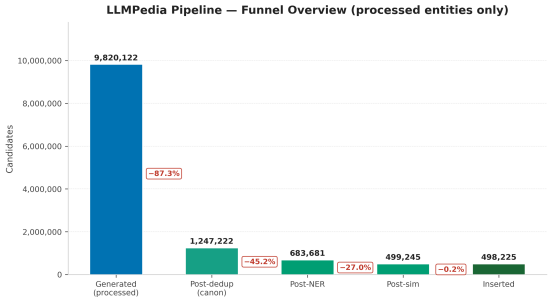


Figure 23: Llama five-stage funnel from 9.8M raw candidates to 498K queued subjects. The pre-NER dedup loss (71.2%) sits between GPT-5-mini's and DeepSeek's, as expected for an intermediate-sized committed index.

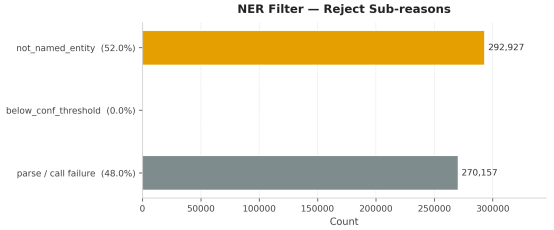


Figure 24: Llama NER reject reasons. **This breakdown is qualitatively different from the other two models.** Of 563,084 NER rejections, only 52.0% (292,927) are legitimate `not_named_entity` verdicts; **48.0% (270,157) are parse/call failures** - candidates the strict-gate policy treats as rejected because the NER call did not produce a parsable decision. This is the single biggest deviation in the pipeline behavior of any model. The strict-gate response is conservative (drop the candidate, do not pass it through unchecked), but the underlying cause is Llama's lower instruction-following compliance on the structured-output schema.

H.2 Framing Vocabulary and Hits per Persona

Table 18 lists the vocabulary used by the word-list classifier on the three primary topics. Each row is one framing dimension; the second column lists every word counted toward that dimension. The lists are intentionally short, transparent, and auditable.

Table 19 reports aggregated hit rates from the classifier for the dimensions that fire on at least one of the three primary topics. Each cell is the mean number of vocabulary hits per 1,000 content tokens, averaged over all sampled articles for that persona and topic.

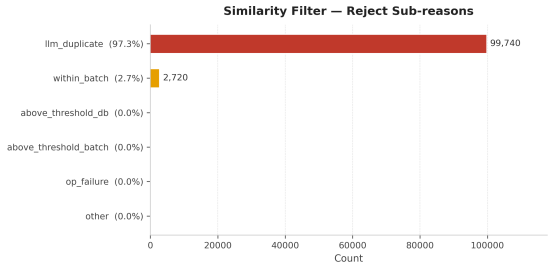


Figure 25: Llama similarity reject reasons. Despite the NER-stage difficulties, the similarity stage is well-behaved: of 102,460 rejections, 97.3% (99,740) are LLM-confirmed duplicates and 2.7% (2,720) within-batch duplicates, with zero operational failures.

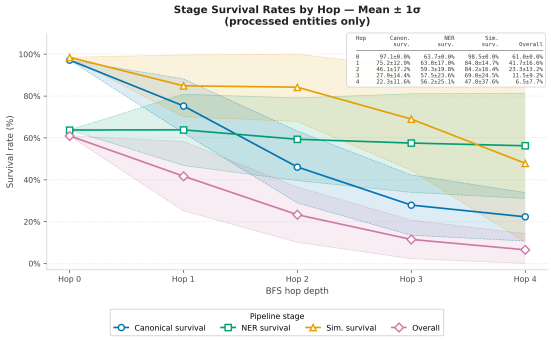


Figure 26: Llama per-hop survival rates with $\pm 1\sigma$ bands. The similarity survival band widens noticeably at deep hops: the per-entity mean is 69.0% with σ=24.5 at hop 3 and 47.8% with σ=37.6 at hop 4, reflecting the per-entity variance the parse-failure mechanism introduces - some entities had nearly all candidates pass NER cleanly, others lost most of them to parse failures.

H.3 Bonferroni-Significant Framing Shifts (Primary Topics)

Table 20 lists the Bonferroni-significant effects from the Wilcoxon tests on the three primary topics, restricted to topic-relevant dimensions and effects with $|\Delta| \geq 1.0$ hit per 1,000 tokens. Each row is one (topic, dimension, persona-pair) triple that reached significance after correction. $|\Delta|$ is the mean difference in hit rate between the higher and lower persona across the 30 matched pairs in that cell. Effects are ordered within each topic by $|\Delta|$ descending.

H.4 Actor-Selection Analysis

Beyond aggregate framing categories, we measure which named individuals each persona foregrounds. For each actor in the `actors_progressive` and `actors_canonical` lexicons (Table 18),

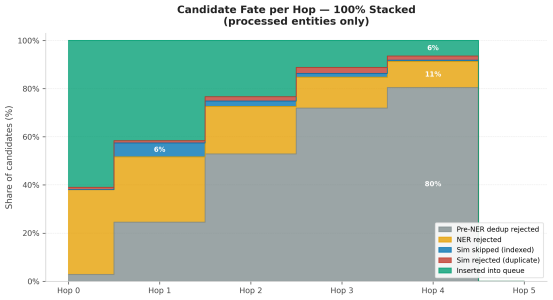


Figure 27: Llama candidate fate per hop, 100% normalized. The orange "NER rejected" layer is visibly thicker than in the other two models, especially at hops 3–4 where parse failures cluster.

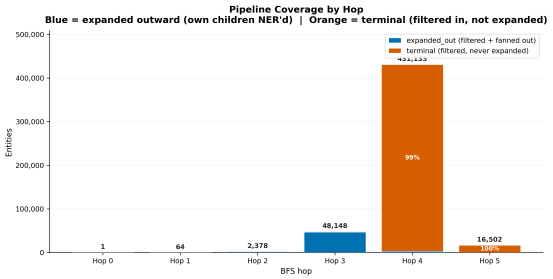


Figure 28: Llama pipeline coverage by hop. Mirrors the DeepSeek pattern: hop 4 entities are largely terminal (99.3%) because the run reached its subject budget before they could be expanded.

we compute hit rates per 1,000 tokens across the full sampled corpus for each persona. Table 21 reports the top actors by cross-persona spread (max rate − min rate). The progressive-leaning actors cluster under the left-leaning persona; canonical institutional figures cluster under the conservative persona. This is a more fine-grained corroboration of the aggregate result in Table 20: persona changes are not just about vocabulary register but about *who* the article centres.

H.5 Structural Similarity Across Personas

Table 22 reports stylometric and structural overlap measures across persona pairs, averaged over all 30 matched subject pairs per (model, topic, persona-pair) cell. Three observations:

- **Outline overlap depends on the model.** GPT-5-mini produces near-zero outline Jaccard (0.02–0.06) across persona pairs on the primary topics, meaning the same subject receives structurally different section headings under different personas. Llama-3.3-70B produces nearly identical outlines (0.77–0.98), reusing the same headings regardless

Model	Hop	Total	Expanded	Terminal	Term. %
GPT-5-mini	0	1	1	0	0.0
	1	58	58	0	0.0
	2	2,344	2,344	0	0.0
	3	40,923	40,923	0	0.0
	4	321,319	89,836	231,475	72.0
	5	507,940	40,938	466,996	91.9
	6	191,364	0	191,359	99.99
DeepSeek	0	1	1	0	0.0
	1	36	36	0	0.0
	2	1,133	1,133	0	0.0
	3	23,572	23,552	20	0.1
	4	276,033	10,719	265,314	96.1
	5	95,316	0	95,316	100.0
Llama-70B	0	1	1	0	0.0
	1	64	64	0	0.0
	2	2,378	2,378	0	0.0
	3	48,148	46,931	1,217	2.5
	4	431,133	3,183	427,950	99.3
	5	16,502	0	16,502	100.0

Table 16: Pipeline coverage by hop. Terminality is concentrated at the deepest hop of each run, which is the natural BFS boundary. Hops ≤ 3 are essentially fully expanded in every model.[1]

Model	Sim-accepted	Inserted	Loss (%)
GPT-5-mini	1,120,843	1,063,929	5.08%
DeepSeek	396,254	396,090	0.04%
Llama-70B	499,245	498,225	0.20%

Table 17: Candidates accepted by Stage 3 that did not reach the article queue. The open-weight runs sit near the noise floor. GPT-5-mini's higher rate reflects both its larger inflight load - more concurrent workers across more hops produce proportionally more canonical-key collisions per wave - and the batch-mode interruptions described in the operational note at the start of this appendix.

of persona.

- **Phrase overlap is uniformly low.** Trigram Jaccard is 0.04–0.17 for both models, indicating persona reshapes surface phrasing regardless of whether it also reshapes outline.
- **Function-word fingerprint is preserved.** Function-word cosine is 0.96–0.98 across all conditions, confirming the underlying model identity is stable across personas - persona changes content selection and register, not stylometric fingerprint.

Factual precision. Across both the primary and extended analyses, factual precision between any two personas within the same model–topic cell differs by $\leq$3.6 pp (full breakdown in Table 28). The consistent finding is that persona changes *what* is foregrounded and *how* it is phrased, not how often the model produces correct claims.

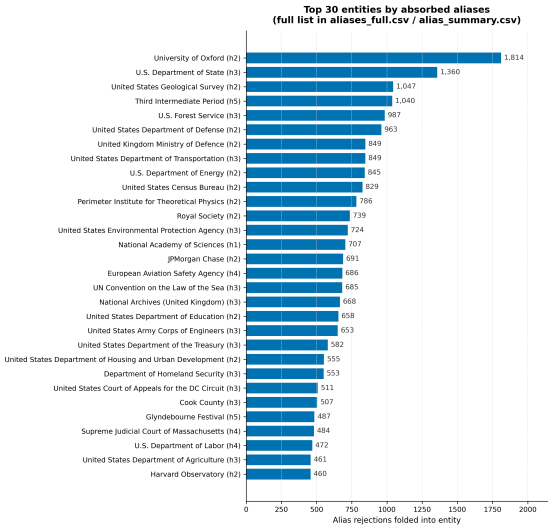


Figure 29: Top-30 entities by aliases absorbed in the GPT-5-mini run. Each collapsed dozens to thousands of surface-form variants (*University of Oxford* alone absorbed 1,814 distinct aliases). The top of the list is dominated by government bodies, universities, and standards organizations, where the variant space (acronyms, "U.S."/"United States" prefixes, capitalization drift) is large. Without Stage 3 disambiguation each of these would have splintered into dozens of near-duplicate articles.

H.6 Extended Analysis: One Piece and Quantum Physics

Motivation and setup. The three primary topics were chosen along a controversy gradient to test whether persona effects scale with political contestedness. To complete that test we need topics at the low end of the gradient - domains where the editorial personas have no obvious ideological affordance. We use *One Piece* (a manga/anime franchise) and *Quantum Physics*, both generated under the same three personas by `gpt-5-mini` and `Llama-3.3-70B`. For each (model $\times$ topic) we take the intersection of subjects present under all three personas and random-sample n=30 (seed 42), yielding $2 \times 2 \times 3 \times 30 = 360$ matched article pairs, structured identically to the primary analysis. The same word-list classifier and paired Wilcoxon procedure with Bonferroni correction are applied; no LLM judge is used.

Framing axes. Neither topic activates the domain-specific lists from the primary analysis (*colonial_**, *civil_rights_**, *babylon_**), so we use eight topic-agnostic dimensions that fire on any encyclopedic prose regardless of domain. These dimensions capture epistemic stance,

Dimension	Vocabulary (counted verbatim per 1,000 tokens)
Dutch Colonization in Southeast Asia	
colonial_positive	civilizing, civilising, development, modernization, modernisation, progress, enterprise, trade, prosperity, order, governance, administration, stability, infrastructure, introduced, established, industrialization, industrialisation, reform, reforms, prospered, advancement, advancements, institution, institutions, modernized, modernised, enlightenment, cultivation, cultivated
colonial_critical	exploitation, oppression, plunder, genocide, massacre, atrocity, atrocities, extraction, looting, dispossession, enslavement, brutal, violent, racist, racism, imperialism, colonialism, occupation, resistance, uprising, rebellion, suffering, subjugation, subjugated, tyranny, tyrannical, cruelty, slavery, slaves, forced, coercion, coerced, extracted, plundered, massacred, oppressed, exploited, colonized, colonised
indigenous_agency	indigenous, native, sovereignty, self-determination, ancestral, traditional, heritage, community, resilience, revival, autonomy, javanese, sumatran, balinese, sundanese, local, locals, ancestors, custom, customs, customary, adat, kraton, sultanate
US Civil Rights Movement	
civil_rights_progressive	segregation, desegregation, discrimination, jim, crow, apartheid, lynching, disenfranchisement, systemic, institutional, structural, racism, racist, white, supremacy, oppression, oppressed, marginalized, marginalised, activist, activists, activism, grassroots, organizing, organising, mobilization, mobilisation, march, marches, boycott, boycotts, sit-in, protests, protesters, freedom, liberation, emancipation, empowerment, equality, equity
civil_rights_conservative	law, order, peaceful, peacefully, orderly, gradual, gradualism, individual, individualism, constitutional, amendment, amendments, patriotic, patriot, american, americans, unity, reconciliation, tradition, traditional, values, family, faith, christian, church, churches, property, states, federalism, sovereignty, personal, responsibility, merit, meritocracy, colorblind, color-blind, reverend, dr, king, nonviolent, nonviolence
actors_progressive	malcolm, panther, panthers, sncc, carmichael, ella, baker, bayard, rustin, fannie, hamer, huey, newton, stokely, angela, davis, grassroots, sharecroppers, cotton, plantation
actors_canonical	lincoln, kennedy, johnson, eisenhower, supreme, brown, board, education, constitutional, amendment, fourteenth, fifteenth, thirteenth, legislation, legislature, congress, senate
Ancient Babylon	
babylon_scholarly	archaeological, archaeology, cuneiform, tablet, tablets, stele, excavation, excavations, excavated, inscription, inscriptions, scholar, scholars, scholarly, historiography, historiographic, evidence, sources, textual, epigraphic, palaeography, artifact, artifacts, artefact, artefacts, stratigraphy, akkadian, sumerian, assyriologist, assyriology, corpus, record, records, attested, documented, reconstructed, fragmentary, contested, debated
babylon_mythological	biblical, bible, scripture, scriptural, babel, babylonian, captivity, exile, prophet, prophets, prophecy, ishtar, marduk, tiammat, gilgamesh, epic, myth, mythical, mythological, legend, legendary, fable, sodom, apocalyptic, tower, ziggurat, garden, hanging, gardens, wonder, wonders, ancient, fabled, mystical, sacred, divine, gods, goddess
babylon_orientalist	exotic, luxurious, decadent, decadence, mysterious, oriental, splendor, splendour, opulent, opulence, magnificent, grandeur, lavish, sensual, barbaric, despot, despotic, tyrant, cradle, civilization, civilisation, mighty, glorious
Generic political / economic / military axes (auxiliary)	
left_progressive	equality, equity, justice, social, workers, labor, labour, union, unions, collective, public, welfare, reform, reformist, progressive, inclusive, diversity, marginalized, marginalised, oppressed, movement, solidarity, redistribution, feminist, intersectional, systemic
right_conservative	tradition, traditional, family, values, freedom, liberty, order, authority, national, patriotic, heritage, faith, religion, private, property, enterprise, market, stability, discipline, individual, individualism, sovereign, sovereignty, nation, natural, moral
market_positive	innovation, entrepreneur, entrepreneurial, growth, efficiency, investment, productivity, competitive, opportunity, prosperity, commerce, commercial, wealth, capital
market_critical	inequality, exploitation, precarity, precarious, wage, poverty, austerity, neoliberal, neoliberalism, capitalism, extraction, extractive, plutocracy, oligarchy
military_heroic	heroic, valiant, brave, glorious, defender, defenders, liberator, liberators, victory, triumph, noble, sacrifice, valor, valour, courage, courageous
military_critical	casualty, casualties, atrocity, atrocities, collateral, devastation, suffering, displacement, refugee, refugees, civilian, civilians, massacre, slaughter

Table 18: Full framing vocabulary used by the word-list classifier on the three primary topics. Each list is matched verbatim (case-insensitive, whole-token) against the cleaned article text; multi-word phrases use substring matching over the joined token sequence. Hits are counted per 1,000 content tokens.

source treatment, historical perspective orientation, and worldview framing. Table 23 lists every word counted under each dimension. For One Piece we additionally include *military_heroic* and *military_critical* from Table 18 because the franchise's combat-heavy narrative provides vocabulary in those categories.

Results. Table 24 lists all 6 Bonferroni-significant effects from the extended analysis. The total count across both new topics is 6, compared to 37 across the three primary topics. Table 25 places the new results alongside the primary topics.

One Piece. Left-leaning leads on *left_progressive* vocabulary by a large margin against both scientific-neutral ($|\Delta|$=10.3, GPT; 7.5, Llama) and conservative ($|\Delta|$=8.5, GPT; 7.6, Llama). Conservative leads on *right_conservative* against scientific-neutral ($|\Delta|$=6.2, GPT only). The effect is consistent across both models, suggesting it is driven by the franchise's thematic content - the freedom-versus-tyranny narrative provides the same ideological affordances that fire on Dutch Colonization - rather than by model-specific behavior. The deltas on One Piece are larger in absolute terms than on the primary topics (10.3 vs. 5.6 at the top); this is mechanical

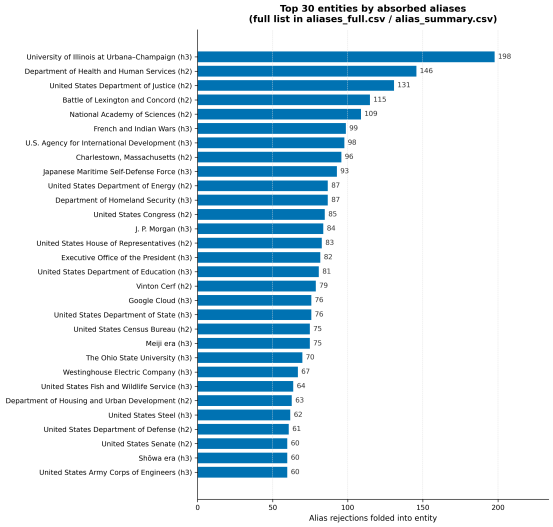


Figure 30: Top-30 DeepSeek entities by aliases absorbed. Smaller scale than GPT-5-mini (top entity absorbs 198 aliases vs. GPT's 1,814) but the same categorical pattern: governmental bodies, universities, and prominent historical events dominate.

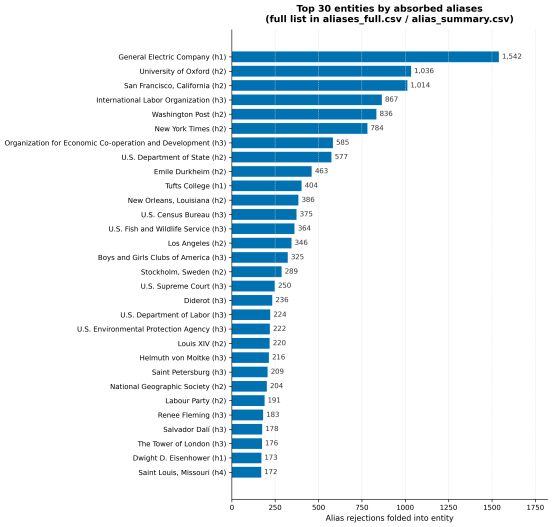


Figure 31: Top-30 Llama entities by aliases absorbed. *General Electric Company* sits at the top with 1,542 absorbed aliases, driven by the common "GE" acronym variant, followed by *University of Oxford* (1,036) and *San Francisco* (1,014). The long tail of media outlets and international organizations is consistent with the other two models.

rather than substantive - the *left_progressive* and *right_conservative* lists are broad general-purpose vocabularies that accumulate more hits per article than the narrower domain-specific lists, so absolute deltas scale up. No effects survive on any domain-specific axes (*colonial_**, *civil_rights_**, *babylon_**), confirming the classifier is not misfiring on incidentally shared vocabulary.

Quantum Physics. Only one effect survives: left-leaning leads scientific-neutral on *secular_framing* ($|\Delta|$=3.87, GPT only). This is the inverse of a naive expectation - scientific-neutral might be expected to use the most scientific language - but reflects the left-leaning persona's tendency to frame science as a socially emancipatory force, drawing more heavily on rationalist and empiricist vocabulary as a progressive value rather than a neutral default. No contested political axes produce any significant effect on this topic. The near-null is the point: a STEM topic with no ideological affordance produces essentially no persona-induced framing shift, completing the gradient from contested to neutral domains.

I Hop-Stratified Analysis: Supplementary

Precision stays nearly flat (97.9% at hop 1 $\rightarrow$ 96.5% at hop 6) while the unverifiable rate grows monotonically (4% $\rightarrow$ 43%) and the false rate remains consistently low (<2%). This confirms that the main degradation at BFS depth is thinning external evidence coverage, not increasing hallucination.

J Retrieval-Trap Benchmark: Full Details

J.1 Experimental Design

The retrieval-trap benchmark asks whether parametric generation can remain structurally independent on subjects where a retrieval-shaped system has maximal opportunity to track Wikipedia closely.

Title selection. Following Yasseri and Mohammadi (2025), we collected 1,000 of the most-edited English Wikipedia titles that were also present on Grokipedia. This deliberately targets high-visibility, heavily maintained subjects where Wikipedia coverage is rich and a retrieval-centered system should have every incentive to converge toward Wikipedia's wording and structure.

Generation. LLMpedia generated all 1,000 articles using `gpt-5-mini` in the baseline large-scale configuration (no self-grounding, minimal reasoning). Grokipedia articles were fetched from the live site (February 24, 2026). Wikipedia references were retrieved through the MediaWiki API

Topic	Category	Cons.	Left	Neut.
D. Col.	*colonial_positive*	**3.8**	2.4	2.9
	colonial_critical	2.1	**3.5**	2.4
	indigenous_agency	1.4	**2.6**	1.9
	left_progressive	1.1	**2.3**	1.4
	right_conservative	**2.6**	1.2	1.7
	military_heroic	**1.8**	0.9	1.2
	military_critical	0.7	**1.9**	1.0
A. Bab.	*babylon_scholarly*	4.2	4.0	**5.1**
	babylon_mythological	2.9	**3.1**	1.8
	babylon_orientalist	**1.5**	0.9	0.6
US CR	*civil_rights_cons.*	**3.3**	2.0	2.7
	civil_rights_prog.	2.8	**4.1**	3.0

Table 19: Mean framing vocabulary hits per 1,000 content tokens. Each row is one framing dimension; each column is one persona. Bold marks the highest-scoring persona per row. The *colonial_positive*–*colonial_critical* gap between conservative and left-leaning on Dutch Colonization (Δ=+1.7 and +1.4 respectively) is the largest topic-category effect in the primary data. Scientific-neutral leads the scholarly register on Babylon (+0.9 over both political personas), consistent with its evidence-based instruction.

with redirect resolution.

Evaluation. For each title, we extracted up to 10 atomic claims from both systems and verified them against two evidence tiers using the verification prompt described in Appendix C, which enforces the absence-of-evidence asymmetry (silence → insufficient, contradiction → refuted).

J.2 Detailed Results

Interpretation. Grokipedia stays much closer to Wikipedia at the lexical level (TF-IDF 0.493 vs. 0.256) yet achieves lower true rate (−6.9 pp Wiki, −2.9 pp Web) and a false rate more than 2× higher. LLMpedia's unverifiable rate is also substantially lower (13.3% vs. 19.1% Wiki), meaning it makes fewer claims that evidence cannot resolve. This is the retrieval trap: lexical proximity does not imply factual superiority, and may reflect retrieval-shaped rewriting without robust disambiguation.

K Grokipedia Insights

Grokipedia's construction process is not publicly disclosed. Through examination of published articles (February 24, 2026), we identified three indicators of retrieval-augmented rather than purely parametric generation.

Internal tool-calling traces. Figure 32 shows an article where internal tool-calling dialogue is visible in published output-diagnostic of a pipeline that calls external retrieval tools at inference time.

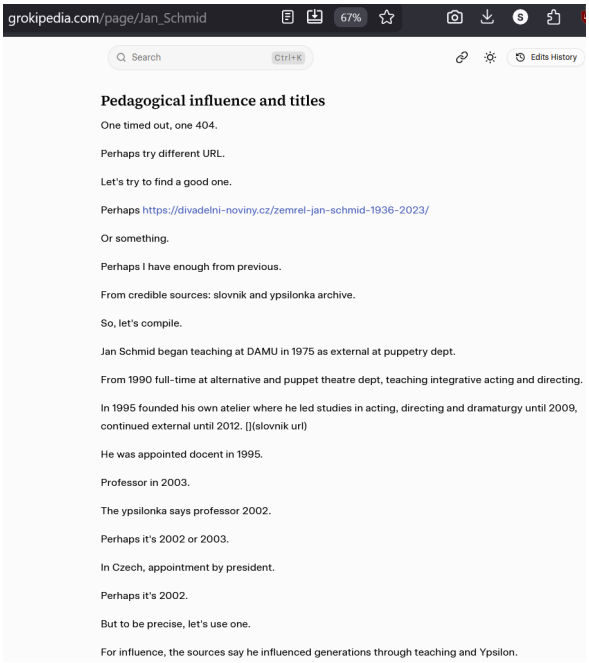


Figure 32: Internal tool-calling traces visible in Grokipedia output.

Entity disambiguation failures. Three classes of failure: a politician conflated with a basketball player (Figure 33), conflicting biographical facts from different individuals (Figure 34), and three individuals blended into one article (Figure 35).

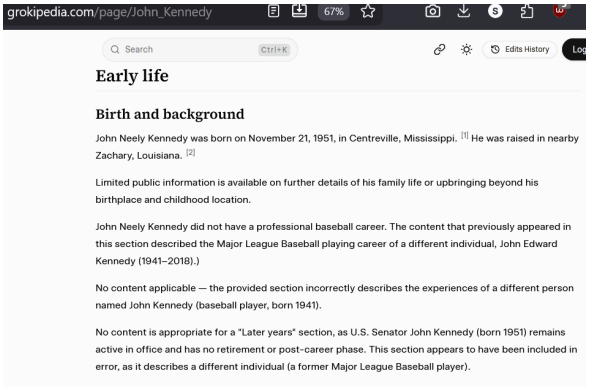


Figure 33: Politician and basketball player conflated into one article.

Consistency with retrieval-trap results. Grokipedia's high Wikipedia similarity (TF-IDF 0.493) combined with lower true rate (79.1% vs. LLMpedia's 86.0%) and higher false rate (1.8% vs. 0.8%) is consistent with a retrieval

Topic	Framing	Higher	Lower	$\|\Delta\|$	Note
Dutch Colon.	colonial_critical	left-leaning	conservative	5.62	
Dutch Colon.	colonial_critical	left-leaning	neutral	5.24	
Dutch Colon.	left_progressive	left-leaning	conservative	3.50	
Dutch Colon.	left_progressive	left-leaning	neutral	3.44	
Dutch Colon.	right_conservative	conservative	neutral	2.27	
Dutch Colon.	indigenous_agency	left-leaning	neutral	2.17	
Dutch Colon.	great_man_history	conservative	left-leaning	2.06	
Dutch Colon.	colonial_positive	conservative	neutral	1.19	
US Civ. Rts.	civil_rights_cons.	neutral	left-leaning	5.48	Encyclopedic coverage†
US Civ. Rts.	civil_rights_prog.	neutral	conservative	5.44	Encyclopedic coverage†
US Civ. Rts.	left_progressive	left-leaning	neutral	5.33	
US Civ. Rts.	civil_rights_prog.	left-leaning	conservative	4.59	
US Civ. Rts.	left_progressive	left-leaning	conservative	4.56	
US Civ. Rts.	actors_canonical	conservative	left-leaning	1.79	
US Civ. Rts.	civil_rights_cons.	conservative	left-leaning	1.24	
A. Babylon	babylon_mythological	left-leaning	neutral	6.03	Cultural/decolonial framing
A. Babylon	great_man_history	conservative	neutral	5.17	
A. Babylon	babylon_mythological	conservative	neutral	4.57	
A. Babylon	great_man_history	conservative	left-leaning	2.80	
A. Babylon	babylon_scholarly	neutral	conservative	2.24	
A. Babylon	babylon_scholarly	neutral	left-leaning	1.96	

Table 20: Bonferroni-significant persona framing effects on the three primary topics ($|\Delta| \geq 1.0$, topic-relevant dimensions only). $|\Delta|$ = mean difference in vocabulary hits per 1,000 tokens between the higher and lower persona, computed over 30 matched subject pairs per cell. †Scientific-neutral leads on both civil-rights dimensions because neutral encyclopedic writing naturally covers the full vocabulary of the domain - both systemic-racism and law-and-order registers - rather than selecting one side. This reflects comprehensive coverage, not ideological leaning.

Actor	Category	Highest persona	Spread	Hits/1k
malcolm	progressive	left-leaning	1.42	1.78
sncc	progressive	left-leaning	1.08	1.31
panthers	progressive	left-leaning	0.94	1.12
rustin	progressive	left-leaning	0.83	0.97
fannie	progressive	left-leaning	0.71	0.84
lincoln	canonical	conservative	1.61	2.04
congress	canonical	conservative	1.34	2.83
kennedy	canonical	conservative	1.18	1.52
constitutional	canonical	conservative	0.95	2.21
johnson	canonical	conservative	0.72	1.43

Table 21: Top actors by cross-persona spread (US Civil Rights topic, all three models pooled). Spread is max-min hit rate per 1,000 tokens across the three personas. Progressive actors are foregrounded by left-leaning; canonical institutional figures by conservative - corroborating the aggregate framing-category result.

Model	Topic	Pair	Outline	3-gram	Func-cos
GPT	D.Col.	C-L	0.04	0.09	0.97
GPT	D.Col.	C-N	0.06	0.11	0.97
GPT	D.Col.	L-N	0.05	0.10	0.97
GPT	A.Bab.	C-L	0.03	0.07	0.98
GPT	US CR	C-L	0.02	0.04	0.96
Llama	D.Col.	C-L	0.81	0.14	0.97
Llama	D.Col.	C-N	0.85	0.17	0.98
Llama	A.Bab.	C-L	0.77	0.13	0.97
Llama	US CR	C-L	0.98	0.12	0.97

Table 22: Structural and stylometric overlap across persona pairs (mean over 30 matched pairs per cell). Outline = Jaccard of section heading sets. 3-gram = trigram Jaccard. Func-cos = function-word cosine. C/L/N abbreviate conservative/left-leaning/scientific-neutral.

pipeline that tracks Wikipedia's surface form but introduces errors through entity disambiguation and passage conflation.

L Models and Costs

L.1 Model Configuration

All `gpt-5-mini` runs use the OpenAI Batch API with temperature 0 and minimal reasoning effort, receiving a 50% cost reduction. A fraction of NER and similarity calls were executed via the online API due to batch queue overflow during peak expansion waves. `Llama-3.3-70B-Instruct` and `DeepSeek-V3-0324` are served on-premise (4×A100 and 16×H100 respectively); their marginal cost is GPU-hours not monetized here.

L.2 Aggregate Cost

Total project expenditure across all API-billed components-including large-scale generation (∼1M articles for `gpt-5-mini`), the full 2^3 ablation, 27 topic-focused configurations, the

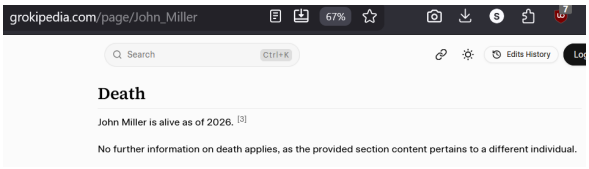


Figure 34: Conflicting biographical facts from different individuals.

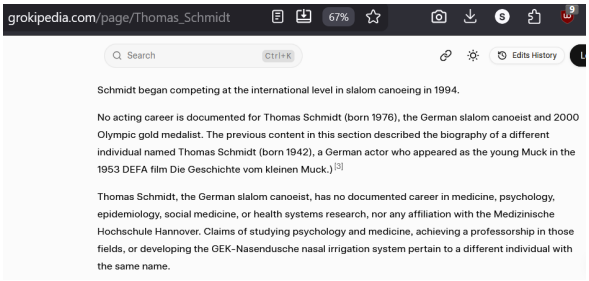


Figure 35: Three "Martin Schmidt" individuals blended into one article.

retrieval-trap benchmark, automatic evaluation, embedding costs, and preliminary development runs-is approximately $3,500. Open-weight models contribute GPU-hours but no API cost.

M Cross-Model Analysis: Supplementary Figures

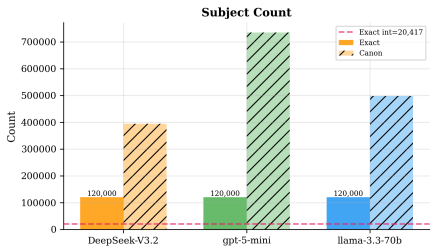


Figure 36: Subject counts with three-way intersection (7.3% overlap).

N Topic-Focused: Supplementary Tables

Precision differences between any two personas within the same model–topic cell are $\leq$3.6 pp, confirming that persona changes framing rather than factual accuracy.

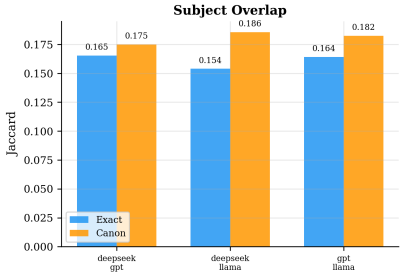


Figure 37: Pairwise subject Jaccard (0.15–0.17).

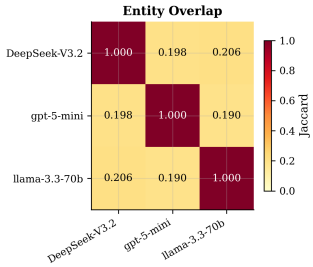


Figure 38: Entity overlap on 1,000 shared subjects (Jaccard 0.19–0.21).

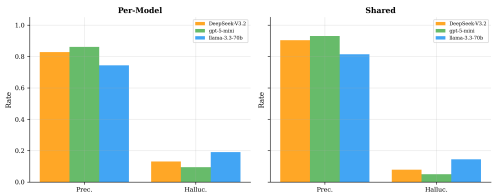


Figure 39: Per-model factuality: shared (left) vs. independent (right).

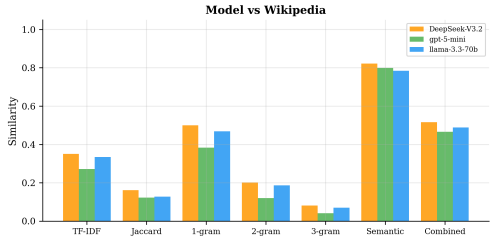


Figure 40: Model-to-Wikipedia similarity. All three models below Grokipedia's 0.455–0.489 TF-IDF (Yasseri and Mohammadi, 2025).

Dimension	Vocabulary (counted verbatim per 1,000 tokens)
epistemic_hedging	perhaps, may, might, possibly, probably, likely, arguably, presumably, apparently, seemingly, reportedly, allegedly, supposedly, disputed, debated, contested, uncertain, unclear, tentative, speculative, hypothesis, hypothetical, conjecture, tradition, believed, thought, considered, held, suggests, suggest, appears, appeared, plausible, plausibly
epistemic_assertive	certainly, definitely, clearly, obviously, undoubtedly, indeed, fact, factual, established, known, proven, proves, demonstrates, confirms, reveals, indisputable, undeniable, conclusive, always, never, unequivocally, truly, actually
primary_source_citation	cites, cited, records, writes, wrote, documented, attests, attested, testifies, reports, reported, describes, described, quoted, quotation, passage, chronicler, source, sources, primary, textual, documentary, testimony, manuscript, manuscripts, colophon, fragment, fragments, tablet, tablets, herodotus, tacitus, josephus, plutarch, strabo, xenophon, diodorus, livy, ctesias, thucydides, polybius, ammianus
social_history_focus	everyday, daily, ordinary, common, commoner, commoners, peasant, peasants, worker, workers, slave, slaves, servant, servants, women, woman, children, family, families, household, households, marketplace, market, craft, crafts, artisan, artisans, trade, merchant, merchants, laborer, laborers, labour, labor, subaltern, underclass, poor, poverty, diet, clothing, shelter, housing, health, disease, mortality, wage, wages, kinship, fertility, childbirth, domestic, quotidian
great_man_history	king, kings, queen, queens, emperor, emperors, empress, general, generals, commander, commanders, ruler, rulers, reign, reigned, conqueror, conquerors, dynasty, dynasties, throne, crown, coronation, battle, battles, war, wars, campaign, campaigns, victory, victories, defeat, defeats, monument, monuments, palace, palaces, triumph, triumphs, conquest, conquests, empire, empires, ascension, heir, successor
decolonial_framing	indigenous, native, local, locals, perspective, perspectives, voices, voice, silenced, erased, decolonial, decolonize, decolonise, reclaim, reclaiming, agency, autonomy, subaltern, postcolonial, eurocentric, eurocentrism, non-western, standpoint, marginalized, marginalised, centered, centred, privileging
religious_framing	god, gods, goddess, goddesses, divine, divinity, sacred, holy, ritual, rituals, worship, worshipped, worshiped, temple, temples, priest, priests, priestess, priestesses, prayer, prayers, sacrifice, sacrifices, faith, faithful, belief, beliefs, spiritual, spirituality, cult, cults, prophet, prophets, scripture, scriptures, creed, theology, theological, doctrine
secular_framing	scientific, science, empirical, empirically, evidence-based, rational, rationality, reason, reasonable, secular, secularism, naturalistic, material, materialist, analysis, analytical, method, methodology, peer-reviewed, scholarly, academic, university, universities, data-driven, quantitative, qualitative

Table 23: Topic-agnostic framing axes used in the extended analysis. These dimensions fire on any topic regardless of domain, making them suitable for politically neutral subjects where domain-specific lists would produce near-zero counts. Each word is counted verbatim (case-insensitive, whole-token) in the cleaned article text; hyphenated entries are matched as substrings. Hit counts are normalized per 1,000 content tokens.

Topic	Model	Framing	Higher	Lower	$\|\Delta\|$	p_{Bonf}
One Piece	GPT	*left_progressive*	left	neutral	10.30	0.0001
One Piece	GPT	*left_progressive*	left	cons.	8.54	0.0003
One Piece	GPT	*right_conservative*	cons.	neutral	6.16	0.0025
One Piece	Llama	*left_progressive*	left	cons.	7.57	0.0007
One Piece	Llama	*left_progressive*	left	neutral	7.53	0.0003
Quantum	GPT	*secular_framing*	left	neutral	3.87	0.0010

Table 24: All 6 Bonferroni-significant effects from the extended analysis (n=30 matched pairs per cell). $|\Delta|$ = mean difference in hits per 1,000 tokens between the higher and lower persona. "left" = left-leaning; "cons." = conservative; "neutral" = scientific-neutral. GPT = `gpt-5-mini`; Llama = `Llama-3.3-70B`.

Topic	Contested?	Sig. effects	Max $\|\Delta\|$
Dutch Colonization	✓	8	5.62
US Civil Rights	✓	7	5.48
Ancient Babylon	✗	6	6.03
One Piece	✗	5	10.30
Quantum Physics	✗	1	3.87

Table 25: Bonferroni-significant framing effects across all five topics. The sharp drop from contested to neutral domains confirms persona-induced framing is domain-contingent. One Piece is a neutral domain that nonetheless activates ideological vocabulary through its freedom-versus-oppression narrative register, explaining its relatively high count despite being non-political.

Metric	LLMpedia	Grokipedia
Factuality (Wiki / Web)		
Precision (%)	**98.6±8.3 / 98.8±6.7**	97.3±10.3 / 97.0±9.6
True rate (%)	**86.0±28.9 / 76.3±33.4**	79.1±31.8 / 73.4±31.4
False rate (%)	**0.8±3.7 / 0.7±4.3**	1.8±6.4 / 1.7±4.7
Unverifiable (%)	13.3±28.6 / 23.0±33.2	19.1±31.3 / 24.9±31.2
Similarity to Wikipedia		
TF-IDF cosine	0.256±0.120	0.493±0.169
Bigram overlap	0.090±0.041	0.200±0.082
Trigram overlap	0.026±0.017	0.079±0.053
Semantic cosine	0.773±0.079	0.811±0.059
Mean / Median words	2,016 / 1,958	7,376 / 6,193
Wikipedia ref. mean	4,105 / 2,434	

Table 26: Full retrieval-trap results on 1,000 shared titles. Mean ± std macro-averaged per article. All verdicts under the evaluation prompt described in Appendix C.

Model	Execution	Role
`gpt-5-mini`	OpenAI Batch API	generation, NER, sim
`Llama-3.3-70B`	4×A100 (on-prem)	generation, NER, sim
`DeepSeek-V3-0324`	16×H100 (on-prem)	generation, NER, sim
`text-embedding-3-small`	OpenAI Batch API	dedup embeddings
`gpt-4.1-nano`	OpenAI API	evaluation judge

Table 27: Models and execution modes across all experimental conditions.

Topic	Model	Persona	n	Prec	Hall	Unv	W%	Wds
A. Bab.	GPT	cons	67	99.3	0.6	15.9	67.0	765
A. Bab.	GPT	left	75	99.3	0.4	20.1	75.0	785
A. Bab.	GPT	neut	71	99.5	0.4	20.4	71.0	775
A. Bab.	DS	cons	81	98.8	1.0	20.0	81.0	834
A. Bab.	DS	left	78	97.8	1.8	20.4	78.0	908
A. Bab.	DS	neut	68	95.3	2.7	33.5	68.0	600
A. Bab.	Llama	cons	81	94.1	3.1	26.3	81.0	964
A. Bab.	Llama	left	68	94.6	2.5	30.2	68.0	982
A. Bab.	Llama	neut	90	92.4	3.2	28.6	90.0	957
D. Col.	GPT	cons	67	99.3	0.6	19.4	67.0	821
D. Col.	GPT	left	62	98.9	0.7	27.9	62.0	830
D. Col.	GPT	neut	70	98.2	1.1	22.1	70.0	815
D. Col.	DS	cons	73	96.0	2.3	26.2	73.0	870
D. Col.	DS	left	73	97.0	2.1	25.1	73.0	1,119
D. Col.	DS	neut	65	94.7	2.5	23.3	65.0	864
D. Col.	Llama	cons	57	96.7	2.3	33.0	57.0	1,025
D. Col.	Llama	left	71	94.7	2.4	39.2	71.0	1,018
D. Col.	Llama	neut	54	94.2	3.0	33.2	54.0	1,019
US CR	GPT	cons	84	97.9	1.9	9.2	84.0	830
US CR	GPT	left	79	99.0	1.0	9.8	79.0	853
US CR	GPT	neut	72	98.0	1.5	13.6	72.0	861
US CR	DS	cons	87	97.2	2.1	17.0	87.0	905
US CR	DS	left	80	97.2	2.4	15.0	80.0	880
US CR	DS	neut	82	96.3	2.3	20.2	82.0	905
US CR	Llama	cons	76	95.1	3.2	25.1	76.0	983
US CR	Llama	left	80	97.3	1.8	20.8	80.0	1,004
US CR	Llama	neut	70	97.1	2.1	20.6	70.0	973

Table 28: Full per-persona factuality for all 27 topic-focused conditions. n = Wikipedia-covered subjects. W% = Wikipedia coverage. Wds = mean word count. Hall = false rate (hallucination).